HuLa App Instructor Guide

2026-10-18 · Hamza Abdelal

How to use this guide

This guide explains the HuLa app (PC version 2.0.5) in plain English for instructors running Hula drone classes. It replaces the official 142-page manual (revised January 2024), which is a rough translation, and the app has no built-in help.

- **Labels in bold** are the words exactly as the app shows them, even when the English is odd, so you can find them on screen.
- Where a label is misleading, the plain meaning follows in brackets, for example **Single driver** (manual flying).
- The Android and iOS apps have the same screens, but the layout differs slightly. Screenshots are from the PC app.
- The **Glossary** at the end lists every confusing label in one place.

Before-class checklist

Most problems in class come from a flat battery, a wrong propeller, or a poor floor. Go through this list the day before.

The day before

- ☐ Charge every battery. The charging box takes about 1 hour; a USB charger takes about 1 h 40 min. On the box, solid red means charging and solid green means full.
- ☐ Use a 5 V 3 A adapter for the charging box.
- ☐ Open the app with internet access and accept any app update (see **Updating firmware and the app**).
- ☐ Download the latest flight firmware and update every drone.
- ☐ Print the QR code cards (0–9) and tag cards you need, and keep their white border clean.

Each drone, before it flies

- ☐ Battery clicked fully in, sticker side down.
- ☐ Propellers on the right motors. Blades marked **75 mmR** go front-left and back-right; blades marked **75 mm** go front-right and back-left. If they are swapped, the drone will not take off.
- ☐ Propeller guards on, and no cracked blades.
- ☐ The downward camera and height sensor on the belly are clean.
- ☐ Battery at least 30% for flips, bounces, circles and other tricks.

The room

- ☐ Indoors, bright, no wind or fans.
- ☐ A floor with visible texture or a printed mat. Plain, shiny, dark or mirror-like floors confuse the downward camera, so the drone drifts.
- ☐ Students stay at least 2 m from a drone during takeoff, flight and landing.
- ☐ Nobody points the laser at eyes.
- ☐ Nobody touches the motors or the heatsink under the drone after flying. They get hot.

Installing the app and logging in

The PC app runs on 64-bit Windows 10 or later only (no Mac). Phones and tablets need Android 11+ or iOS 15+.

Device	Where to get it
Windows PC	download.hg-fly.net/app/hula_pc.html
Android phone or tablet	download.hg-fly.net/app/hula_app.html
iPhone or iPad	App Store, via the same link

PC install tips

- Install into a folder whose path has only English letters. A Chinese or Arabic folder name can stop the app starting.
- If the installer fails, turn off the Windows firewall during installation, then turn it back on.

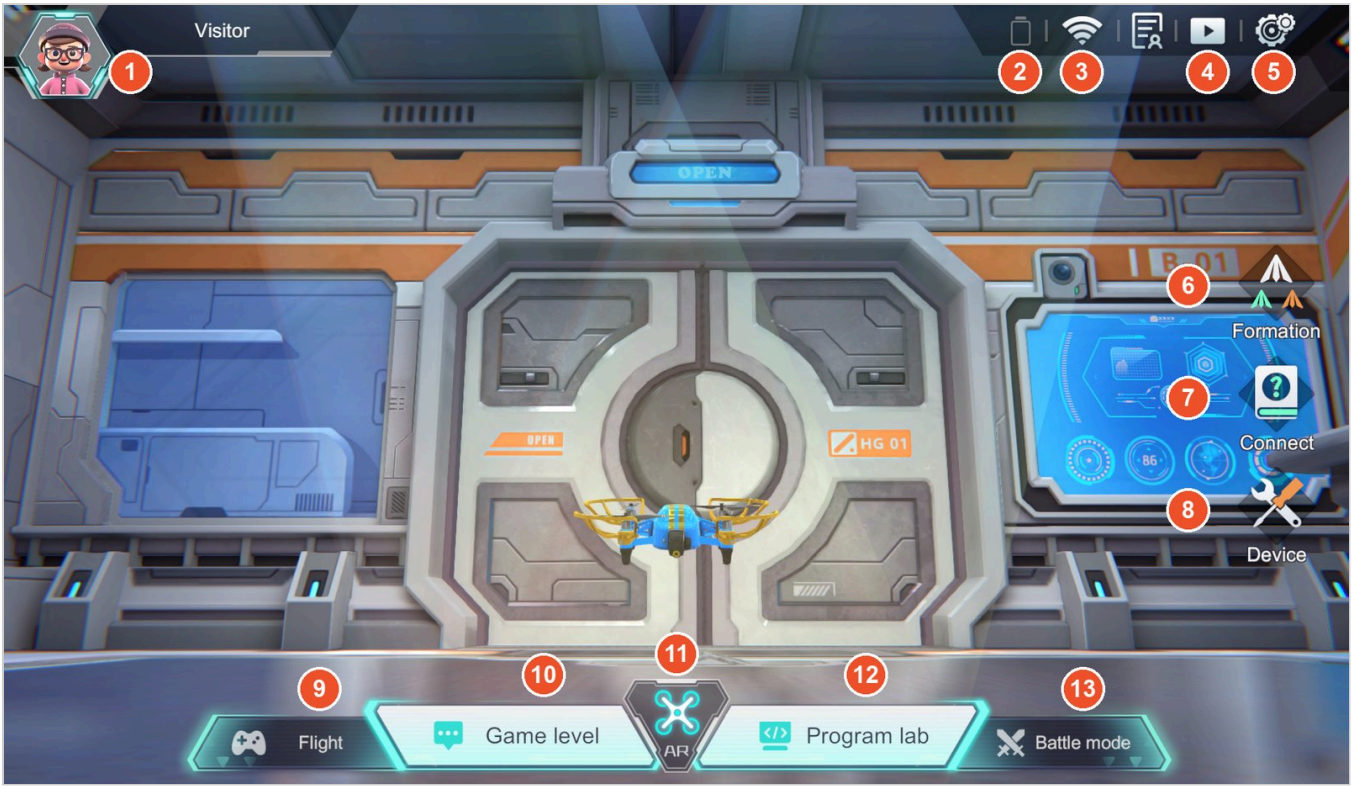
Logging in

- Pick **English** on the login screen. Accounts on the Chinese server and the international server are separate, so a student who registered in one language won't find their work in the other.
- **Visitor login** needs no account and is fine for flying. Visitor programs can't be saved to the cloud, so use real accounts for anything students must keep.
- Registering needs an email address (or a Chinese phone number) and a verification code.
- A login stays valid for 30 days.

Classroom tip: one account per drone or per team works better than one per student. Work is tied to the account, so students can pick up where they left off.

Home screen

The home screen has five big buttons along the bottom, status icons along the top, and three shortcuts on the right. The numbers match the table below.



#	Button	What it really does
1	**Profile** (your picture)	Name, password, messages, technical support. Shows "Visitor" when logged in as a visitor.
2	Battery	The connected drone's battery. Empty when no drone is connected.
3	WiFi	Shows whether you're connected to a drone; tap to open your device's Wi-Fi settings
4	**Album** (play icon)	Photos and videos stored on the drone
5	**Settings** (gear)	Video, Control, System and Firmware settings
6	**Formation**	Multi-drone formation shows (several drones flying a routine together). Needs router mode.
7	**Connect**	Explains the two connection modes, AP and STATION (see **Connecting to the drone**)
8	**Device**	The list of drones the app can see; choose which one is yours for battles. Opens only with a drone connected.
9	**Flight**	Manual flying with on-screen joysticks (the manual calls it **Single driver**)
10	**Game level**	Programming puzzles on a 3D map, no drone needed
11	**AR**	Augmented-reality mode. The official manual doesn't cover it.
12	**Program lab**	The block programming editor that flies the real drone
13	**Battle mode**	Laser battle games between drones

The unnumbered icon between WiFi and Album (a page with a person) did nothing when logged in as a visitor.

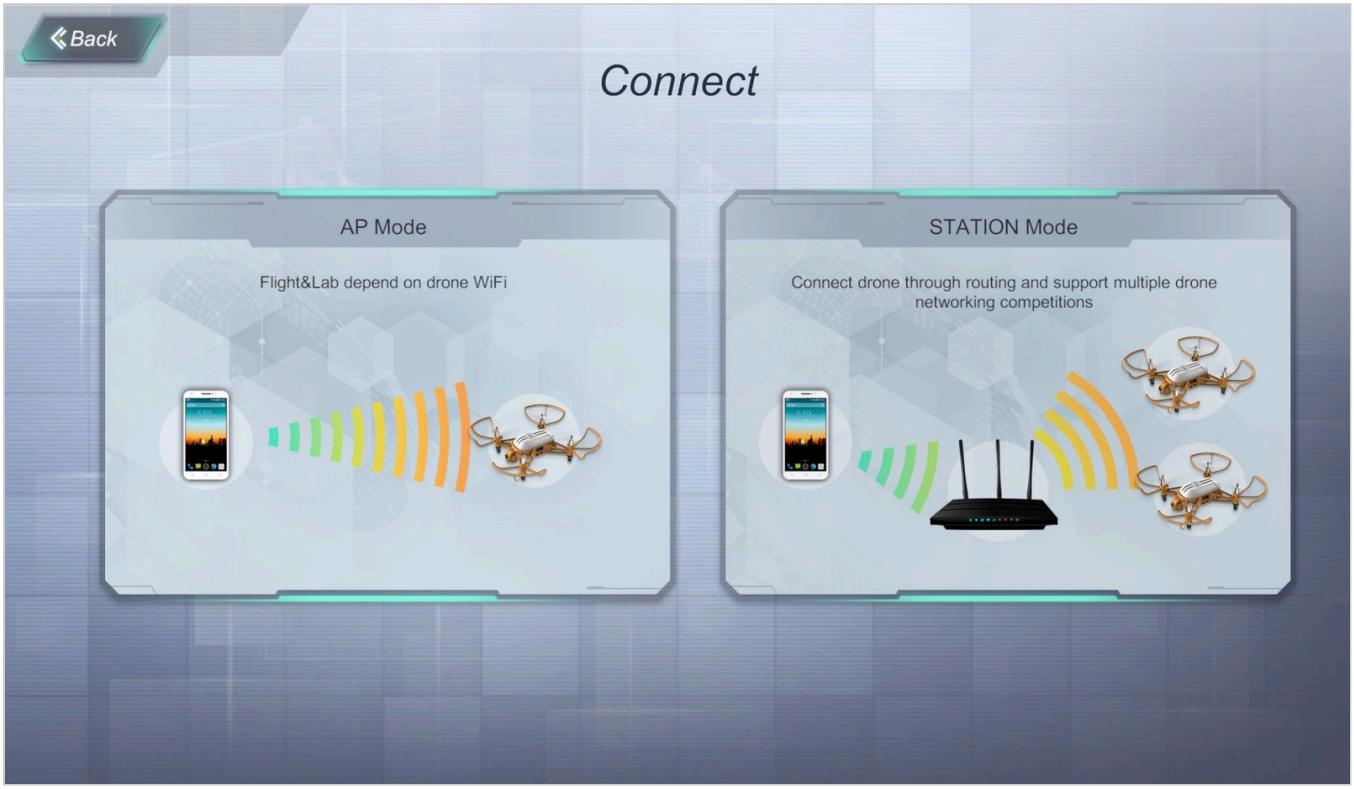
The login screen comes first. Choose **English** (top right), tick **I agree with Agreement & Privacy Policy**, then log in or tap **Visitor**.



Connecting to the drone

The drone has two connection modes. **Direct mode** (one device, one drone) is what you use for flying and programming. **Router mode** puts many drones on one Wi-Fi router, and is needed only for team battles (Scrimmage). New drones start in direct mode.

The app's **Connect** screen calls them **AP Mode** (direct) and **STATION Mode** (router). The manual also says "direct connection mode" and "networking mode". They are the same two modes.



Which mode is it in? Watch the light after power-on. A purple flash means direct mode; a white flash means router mode. Press the power button 3 times quickly to switch.

Direct mode (normal classes)

1. Hold the power button on the back for 2 seconds. The light flashes red, green and blue (self-check), then purple.
2. On the PC or tablet, join the Wi-Fi named **Hula-XXXXXXXX** (the X's are the drone's serial). The password is **12345678**.
3. Open the app. When the drone's light turns solid green, it is connected.

If the Hula Wi-Fi doesn't show up: refresh the Wi-Fi list, restart the drone, or press the power button 3 times to make sure it is in direct mode.

Classroom tip: label each drone and its laptop with the same number, matching the last digits of the Hula-XXXXXXXX name, so nobody connects to a neighbour's drone.

Router mode (team battles)

1. Set your router's Wi-Fi name to **Hula-Battle** and password to **HG888888** (the drones' factory setting). Alternatively, connect each drone directly, use **Settings → System → Modify Drone WiFi** to send it your router's name and password, then switch it to router mode.
2. Switch each drone to router mode (power button 3 times; white flash).
3. Connect every PC or tablet to the same router.
4. The first time, open **Device** on the home screen, select your drone (it flashes yellow), check the serial number (SN), and tap **Battle** to claim it.

What the drone's light means

Light	Meaning
Red, green, blue flashing	Self-check after power-on
Purple flash	Switched to direct mode
White flash	Switched to router mode (red for 3 s means it couldn't reach the router)
Green breathing	Direct mode, waiting for the app
White breathing	Router mode, waiting for the app
Solid green / white	Connected to the app
Blue double flash	Obstacle detected by infrared
Orange breathing	In a special flight mode (follow, line patrol, circle)
Orange flashing / solid 5 s	Looking for a target / target recognised
Red slow flash	Low battery
Solid red	Hardware fault or failed self-check
Blue flashing	Firmware being uploaded
Green fast flash after reset	Wi-Fi reset to factory

Resetting the drone

There is a small reset hole next to the power button. Use a pin.

- **Hold 5 s:** resets the drone's Wi-Fi to factory settings (Hula-XXXXXXXX, password 12345678). The green light flashes fast when done.
- **Hold 10 s:** rolls the flight firmware back to the previous version. The drone restarts when done.

Updating firmware and the app

A firmware update has two parts: download it on the internet, then install it while connected to the drone. The laptop can't be on the internet and the drone's Wi-Fi at the same time, so do the download first.

App update

1. Open the app while on the internet. If a new version exists, a pop-up offers **Upgrade Now**. You can also check under **Settings → System → App Version**.
2. Let it download and follow the installer. Don't close the app or lose Wi-Fi during the download.

Drone (flight control) firmware update

1. **On the internet:** go to **Settings → Firmware → Local File** and download the latest firmware to the laptop.
2. **Connect to the drone** (direct mode). **In Settings → Firmware, the Pilot Version row now offers the upgrade (the manual calls it Upgrade Flight Control)**. Tap it.
3. The firmware uploads while the drone's light flashes blue. Then the drone installs it by itself for a few minutes: lights go off, it does a red-green-blue self-check, and the camera moves.
4. The pop-up closes on its own. Reconnect to the drone's Wi-Fi; it should say **Already the latest version**.
5. If it fails, tap **Re upgrade**.

During any update: don't power off the drone, close the app, or disconnect Wi-Fi. A broken firmware update can damage the drone. If a new firmware causes problems, holding the reset button for 10 s rolls back to the previous version.

Classroom tip: download the firmware once, then update all drones one after another from the same laptop.

Flying manually (Flight / Single driver)

Tap **Flight** on the home screen to fly with on-screen controls. This is the best first lesson: students feel how the drone moves before they program it.

Screen layout



Without a drone connected, the background is a 3D hangar. Once connected, it shows the drone's live camera.

#	Control	What it does
1	Back	Return to the home screen. Land first.
2	**STOP**	Emergency stop.
3	Drone status	Red X = no drone connected. Tap for the status panel when connected.
4	Battery	Drone battery level
5	WiFi	Connection to the drone
6	Help (?)	Help for this screen
7	Settings	Same Settings as the home screen
8	Video / photo switch	Choose camera or video mode for the shutter button
9	Shutter (white circle)	Take a photo, or start/stop video. Video stops at 8 minutes.
10	Left joystick	Up/down arrows climb and descend; curved arrows turn left and right ("American hand" layout)
11	Right joystick	Fly forward, back, left, right
12	Gimbal slider	Tilt the front camera up and down
13	Takeoff / Landing	Hold 1 s to take off to 1 m. Tap again to land, then confirm.
14	Album	Photos and videos on the drone

#	Control	What it does
15	Advanced functions	The one-tap tricks below. Opens only with a drone connected.

To swap the joysticks, change **Control Mode** to **Japanese hand** in **Settings → Control**.

One-tap tricks (Advanced functions)

They need 30% battery or more and at least 0.5 m height. To pause any trick, hold the joystick in any direction for 1 second.

Trick	What happens
Follow	Follows a QR code (0–9) held by a person, or a colour. Hold the code 1–2.2 m away and move it slowly.
Line patrol	Follows a line on the floor. Hover over the line, then tap start. Same line rules as the **Line patrol** block.
Draw a vertical circle	Flies a vertical loop centred 0.75 m above the drone.
One click bounce	Bounces up and down once.
One click fly around	Orbits a point 1.5 m in front of the nose.
Circle	Spins once clockwise on the spot.
Curve	Curves clockwise to a point 1.5 m ahead and to the right.
One click flip	Flips in the direction you swipe.

Aircraft status panel (tap the health dot)

Shows firmware version, battery, signal, IMU (motion sensor), laser receiver, optical flow (downward camera), total flight hours, temperature, storage and ToF (height sensor). ToF and optical flow showing **red while the drone sits on the floor is normal**, because the ground is too close for them. They turn green after takeoff.

Safety

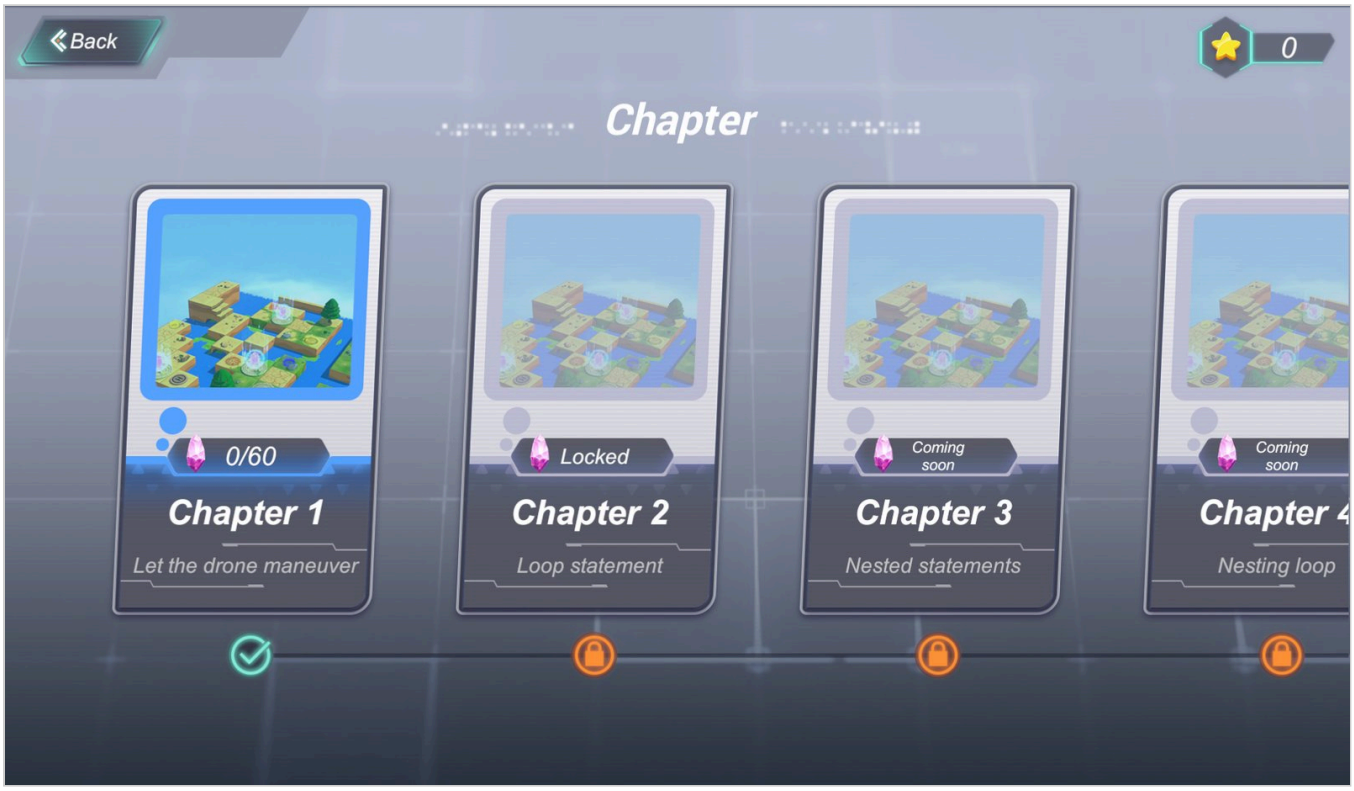
- Height for best control: 1–4 m. Never below 0.5 m, where it can crash; the maximum is 10 m.
- Keep the app on the flying screen. Leaving the screen, closing the app or losing Wi-Fi in flight is risky. If the connection is lost, the drone lands by itself after 10 seconds.
- Never launch from a hand. Stay 2 m away.
- Replace the battery after about 30 hours of total flight time (shown in the status panel).

Game levels (programming puzzles)

Game level on the home screen is a set of block-programming puzzles played on a 3D map. No drone is needed, so it suits the first programming lesson or students waiting for a drone.

- Levels are numbered chapter-level: **1-2** is chapter 1, level 2.
- Drag blocks into the programming area and press **Run** to watch the drone on the map. Collect enough items to pass the level perfectly.
- The reminder (lightbulb) button explains the level's goal. **Play multiple** speeds the replay up to 3×. **Play again** resets blocks and map.
- Pinch or scroll to zoom the map, drag to rotate the view.
- Stay online while playing, or level progress may not be saved.

What's in version 2.0.5: Chapter 1 "Let the drone maneuver" is open, with 60 stars to earn. Chapter 2 "Loop statement" unlocks after it. Chapters 3 "Nested statements" and 4 "Nesting loop" are marked **Coming soon**.





Inside a level, only three block categories are offered: **Aircraft**, **Interaction** and **Control**. The top bar shows how many blocks you've used against the best solution ("block num 1 (bestest 5)"), plus **Clear** to empty the area. Top right are **Tips**, playback speed (**x1**) and **Retry**. The **Mission** panel lists what to collect (crystals) and reach. Press the play button at the bottom right to run.



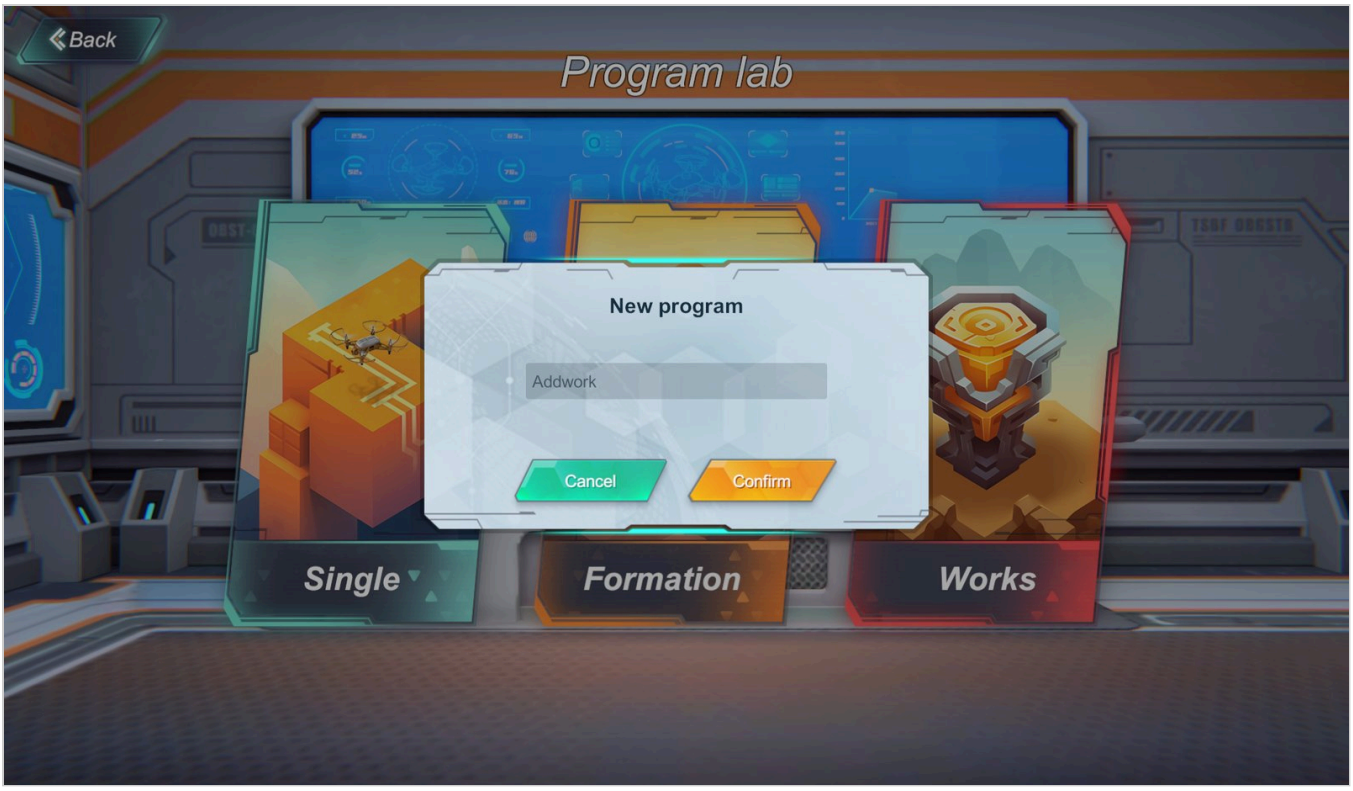
Programming lab

Program lab on the home screen opens the block editor that flies the real drone. Students build a program, test it on the 3D simulator (**SIM run**), then fly it for real.

Program lab first shows three cards:

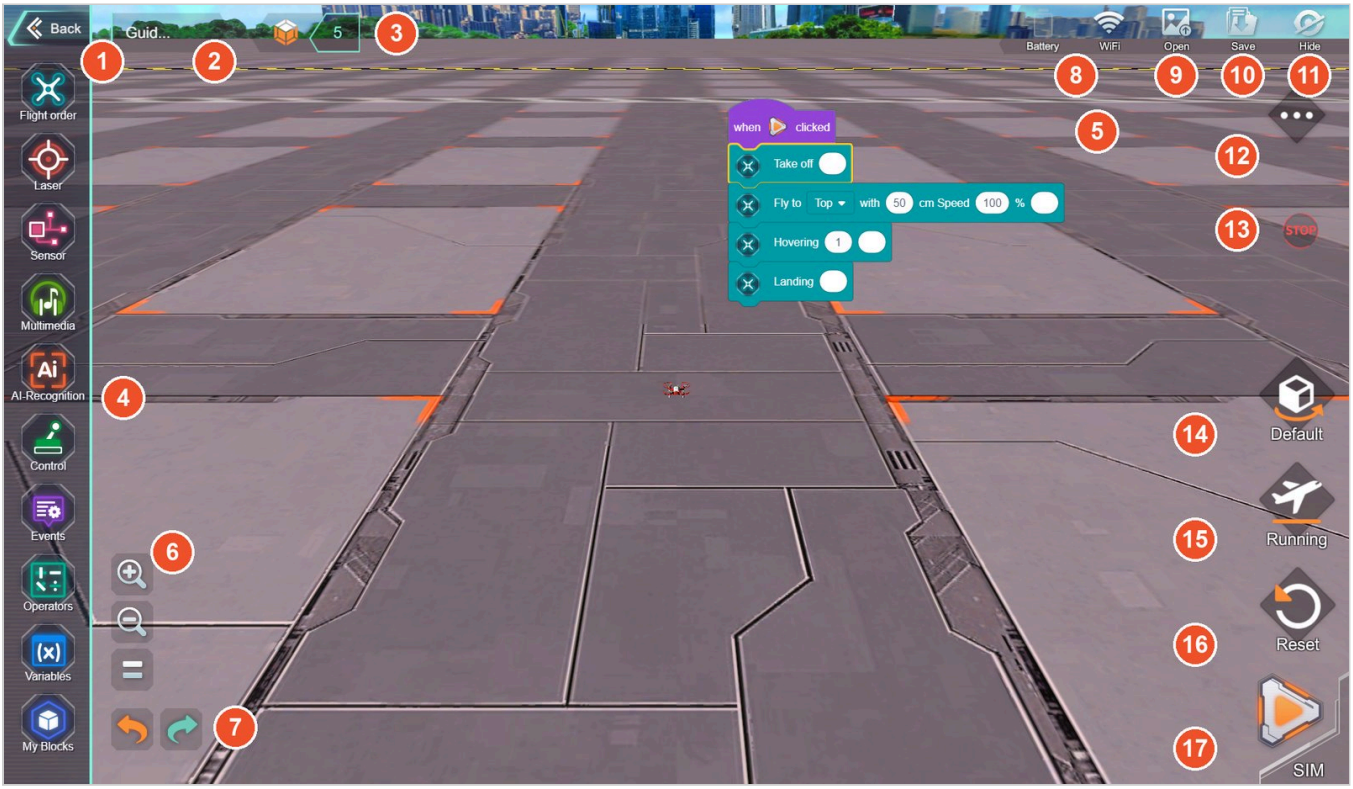
- **Single:** program one drone. This is what classes use. It asks for a program name, then opens the editor.
- **Formation:** program several drones together (router mode).
- **Works:** saved programs. In version 2.0.5 it would not open while logged in as a visitor, so use real accounts if students need to keep their work.





In the Programming lab the drone starts with its **light off and obstacle avoidance off**. Turn them on with blocks if you need them. Indoors only.

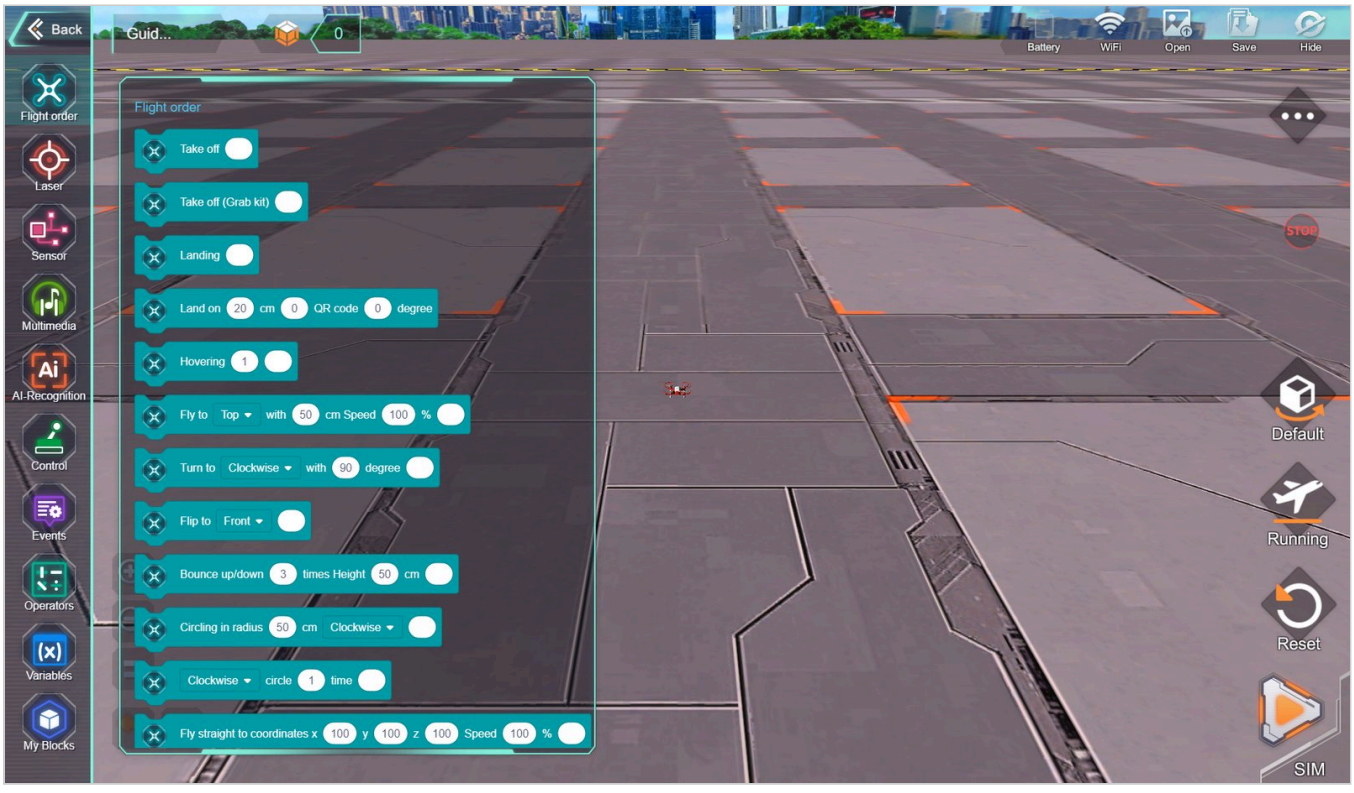
Editor layout



#	Part	What it does
1	Back	Leave the editor. It asks **Save** or **Exit** (exit without saving).

#	Part	What it does
2	Program name	The name typed when the program was created
3	Block counter	How many blocks the program uses
4	Category buttons	Flight order, Laser, Sensor, Multimedia, AI-Recognition, Control, Events, Operators, Variables, My Blocks. Click one to open its block palette.
5	Programming area	Drag blocks here and snap them together. The 3D simulator stage is behind them.
6	Zoom in / Zoom out / "="	Change block size; "=" re-centres the blocks
7	Undo / Redo	Step back or forward
8	Battery, WiFi	Drone battery and connection
9	**Open**	Turns the live camera picture (image transmission) on. Needed by the AI blocks that run in the app.
10	**Save**	Save the program
11	**Hide**	Hide the blocks so you can look around the simulator
12	More (•••)	Day/night mode and grid lines on the simulator
13	**STOP**	Emergency stop: the real drone lands where it is
14	**Default** (view)	Simulator camera: roaming, tracking (default), first-person or overhead
15	**Running**	**Press and hold** to fly the real drone: it takes off and runs the program
16	**Reset**	Put the simulator and program back to the start
17	**SIM**	Run the program on the 3D drone on screen

Clicking a category opens its palette over the stage:



Basic moves

1. Drag a block from the palette into the programming area. It snaps onto the block above when you let go close to it.
2. To delete a block, drag it back onto the palette on the left.
3. Start every program with **when clicked** (Events), then **Take off**, and end it with **Landing**.
4. **SIM run** first. If the 3D drone does what you expect, connect the real drone and **hold Running**.
5. **Stop** is the emergency button: the drone lands where it is.

Coordinates: at takeoff the drone sets its starting point as (0, 0, 0). If it takes off from a table and flies lower than the table top, heights go negative. What matters for safety is the real height above the floor: below 0.5 m the drone may crash.

Saving and managing work (My work)

- Click a saved program to open and edit it.
- The "…" button on a program uploads it to the cloud, or renames, copies or deletes it.
- **Sync** downloads all cloud programs to this device. Upload local changes first, or the cloud copy overwrites them.
- On PC, **Export** saves selected programs to files and **Import** loads them. Use this to share a starter program with every laptop.
- Programs live on the device until uploaded. Back up before reinstalling or swapping laptops.

Block reference

The Programming lab has 102 blocks in 10 categories. Each entry below shows the block as it looks in the app, then what it does, its inputs, what it gives back, and classroom notes. Allowed values come from the app's own code (version 2.0.5). Where they differ from the official manual, the code wins.

How to read a block entry

Block shapes

Shape	Name in this guide	What it means
Rounded top, bump underneath	**Start block**	Starts a script. Nothing goes above it.
Notch on top, bump underneath	**Command**	Does one action. Snaps into a stack and runs top to bottom.
C shape	**Wrapper**	Holds other blocks inside it (loops and ifs).
Oval	**Value**	Gives back a number or text. Drop it into a white slot of another block.
Hexagon	**True/false**	Gives back true or false. Drop it into an **if**, **wait until** or **repeat until**.

The empty white oval at the end of a flight block is an attachment slot. It is empty by default and the block works without it. Drop an oval **Light - Color value** block into it to light the drone during that move, or an oval **Lasing** block to fire the laser during that move. The attachment stops when the move ends.

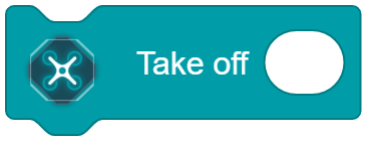
Typing numbers: if you type a value outside the allowed range, the app refuses it and keeps the old value. Nothing on screen says why.

SIM run: the Programming lab can run a program on a 3D drone on screen before the real flight. Blocks that need the real drone's sensors or camera do nothing in SIM run. Those entries say **SIM run: no**.

Flight order

Teal blocks that move the drone. Directions are always relative to where the drone's nose points at that moment. 100% speed = 1 m/s.

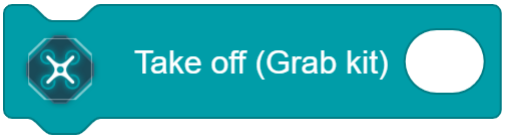
Take off



Starts the motors, climbs to 1 m (100 cm) and hovers. Every flight program needs it before any other flight block, or nothing will fly.

- **Shape:** Command
- **Inputs:** attachment slot (optional): a light or lasing value block
- **Gives back:** nothing
- **SIM run:** yes
- **Notes:** students stay 2 m back. Below 0.5 m the drone can crash, so don't program moves that take it lower.

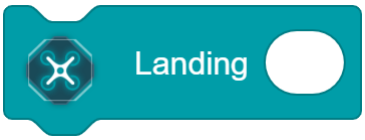
Take off (Grab kit)



The same as **Take off**, for a drone fitted with the clamp (grab) add-on kit.

- **Shape:** Command
- **Inputs:** attachment slot (optional)
- **Gives back:** nothing
- **Notes:** without the kit fitted, use plain **Take off**.

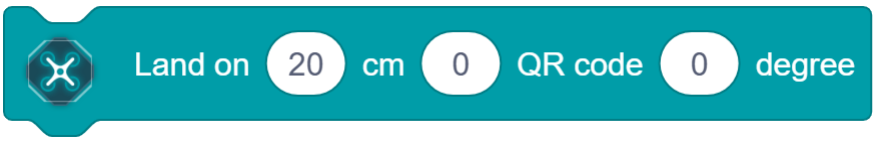
Landing



Lands straight down where the drone is and stops the motors. Put it at the end of every flight program, or the drone keeps hovering after the program ends.

- **Shape:** Command
- **Inputs:** attachment slot (optional)
- **Gives back:** nothing
- **SIM run:** yes

Land on QR code

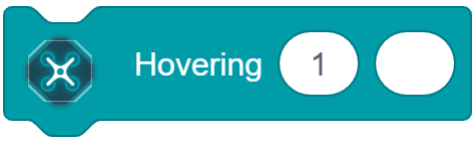


The downward camera looks for a numbered QR code on the floor. The drone centres over it, turns to the chosen angle and lands on it.

Input	Meaning	Allowed	Default
first number (cm)	Printed size of the QR code. Measure one side of the black square.	6–30 cm in practice	20
second number	Which QR code to land on	0–9	0
degree	Extra turn after lining up with the code	−180 to 180	0

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** no
- **Notes:** the code must be under the drone. At 1 m height it must be within a 30 × 30 cm square below; at 2 m, 60 × 60 cm; at 3 m, 100 × 100 cm. Above 3.5 m it won't see the code. Use good light, and don't cover the white border.

Hovering

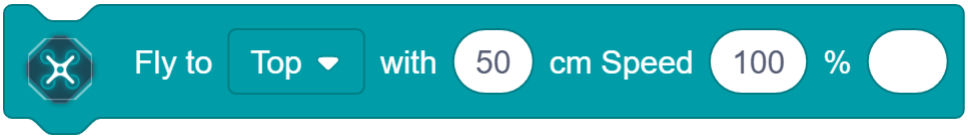


Holds the drone still in the air for a number of seconds.

Input	Meaning	Allowed	Default
number	Seconds to hover	1–600	1
attachment slot	Light or lasing while hovering	optional	empty

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** yes
- **Notes:** add 1–2 s of hovering after each move so the drone settles. Most "wobbly" programs are missing these pauses.

Fly to

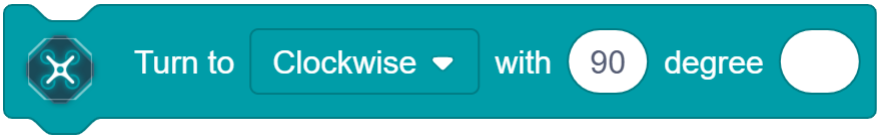


Flies in a straight line in one direction for a set distance.

Input	Meaning	Allowed	Default
direction	Top (up), Bottom (down), Front, Back, Left, Right	dropdown	Top
cm	Distance	5–500 cm	50
Speed %	Speed; 100% = 1 m/s	1–100	100
attachment slot	Light or lasing during the move	optional	empty

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** yes
- **Notes:** this is the block students use most. For accurate track flying, use 30–50% speed.

Turn to



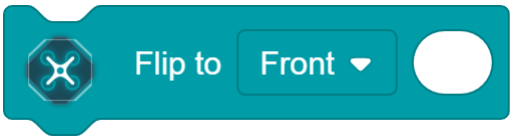
Rotates the drone on the spot (yaw). It does not move forward.

Input	Meaning	Allowed	Default
direction	Clockwise (right) or Anticlockwise (left)	dropdown	Clockwise

Input	Meaning	Allowed	Default
degree	How far to turn	−360 to 360	90
attachment slot	Light or lasing while turning	optional	empty

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** yes
- **Notes:** after a turn, "Front" in later blocks means the new nose direction. Use the dropdown for direction, not negative numbers.

Flip to

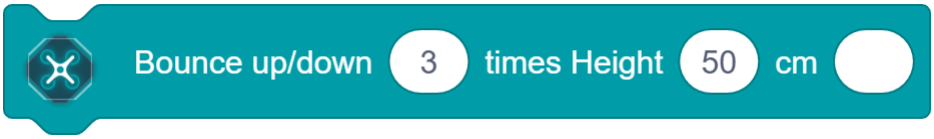


Does a fast 360° flip in the chosen direction.

Input	Meaning	Allowed	Default
direction	Front, Back, Left, Right	dropdown	Front
attachment slot	Light or lasing during the flip	optional	empty

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** yes
- **Notes:** battery must be over 30%. Fly 1–3 m high, indoors, away from people. The drone drops a little during the flip.

Bounce up/down

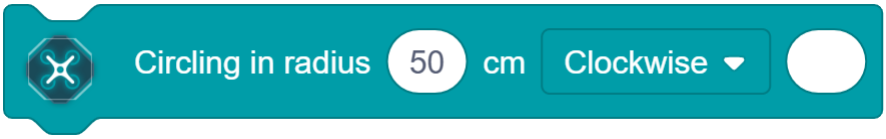


Bounces the drone up and down several times, fast (about 1.5 m/s).

Input	Meaning	Allowed	Default
times	Number of bounces	1–10	3
Height cm	Size of each bounce	20–200 cm	50
attachment slot	Light or lasing while bouncing	optional	empty

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** yes
- **Notes:** start from at least 1 m high. Battery over 30%.

Circling in radius

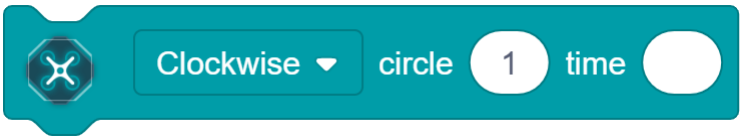


Flies one full circle around a point in front of the drone. The nose keeps facing the centre.

Input	Meaning	Allowed	Default
cm	Distance from the drone to the centre point (in front of the nose)	15–300 cm	50
direction	Clockwise or Anticlockwise	dropdown	Clockwise
attachment slot	Light or lasing while circling	optional	empty

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** yes
- **Notes:** a 150 cm radius needs a 3 m clear circle. Battery over 30%.

Circle (spin on the spot)

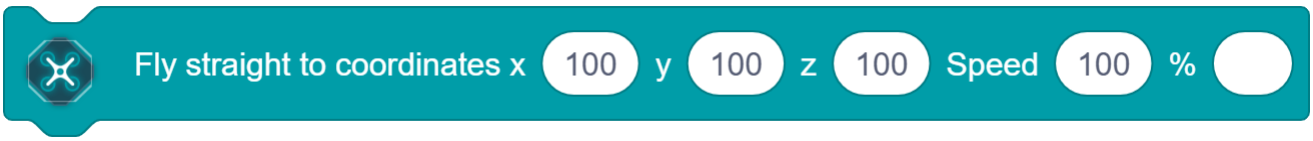


Spins the drone a number of full turns where it is. Despite the name, it does not fly in a circle. That is **Circling in radius**.

Input	Meaning	Allowed	Default
direction	Clockwise or Anticlockwise	dropdown	Clockwise
time	Number of full turns	1–10	1
attachment slot	Light or lasing while spinning	optional	empty

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** yes

Fly straight to coordinates



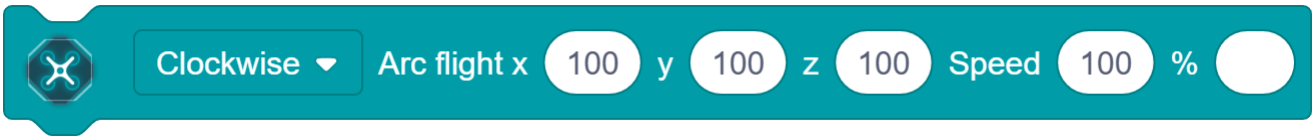
Flies in a straight line to a point given as x, y, z in centimetres from where the drone is now. +x is right, +y is forward (nose), +z is up.

Input	Meaning	Allowed	Default
x	Right (+) or left (−)	−1000 to 1500 cm	100

Input	Meaning	Allowed	Default
y	Forward (+) or back (−)	−1000 to 1500 cm	100
z	Up (+) or down (−)	−300 to 400 cm	100
Speed %	100% = 1 m/s	1–100	100
attachment slot	Light or lasing during the move	optional	empty

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** yes
- **Notes:** each move counts from where the previous one ended. To go back, use the negative numbers. One move is capped at 500 cm in total; split longer moves into two blocks. With **Open QR code carpet positioning** turned on, x, y, z become fixed positions on the QR mat instead.

Arc flight

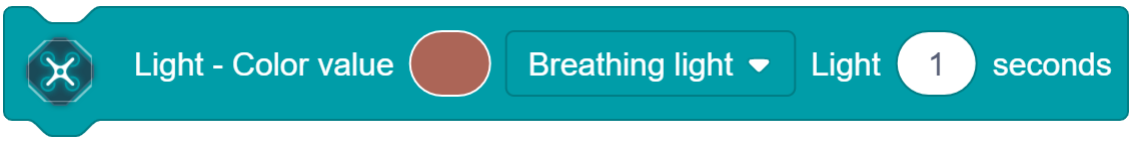


Flies to a point along a curved path instead of a straight line. Inputs work like **Fly straight to coordinates**.

Input	Meaning	Allowed	Default
direction	Curve clockwise or anticlockwise	dropdown	Clockwise
x, y, z	Target point, cm from where the drone is now	x, y: −1000 to 1500; z: −300 to 400	100 each
Speed %	100% = 1 m/s	1–100	100
attachment slot	Light or lasing during the move	optional	empty

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** yes
- **Notes:** the path is half an ellipse with the start and end points as its long axis. Keep the height under 3 m. Same 500 cm cap per move.

Light - Color value (command)



Turns on the drone's top light in a colour and effect for a number of seconds.

Input	Meaning	Allowed	Default
colour circle	Click to pick the colour	colour picker	random
effect	Breathing light (slow fade), Crush out (light off), Steady on, Colorful lantern (cycles 7 colours), Flash (fast blink)	dropdown	Breathing light

Input	Meaning	Allowed	Default
seconds	How long the light stays on	1–100	1

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** yes
- **Notes:** the program does not wait for the light. A new light block replaces the current one at once, so with several in a row only the last shows. Put **Hovering** or **wait** between them. "Crush out" means turn the light off.

Light - Color value (attachment)

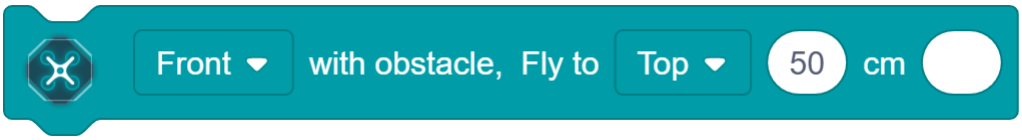


The oval version of the light block. It fits only in a flight block's white attachment slot. The light stays on for as long as that move takes.

Input	Meaning	Allowed	Default
colour circle	Colour	colour picker	random
effect	Same effects as above	dropdown	Breathing light

- **Shape:** Value (attachment)
- **Gives back:** a light setting for the flight block it sits in
- **Notes:** a good way to show which leg of a track the drone is on.

Obstacle, then fly



If the infrared sensors detect an obstacle in one direction, the drone moves a set distance in another direction.

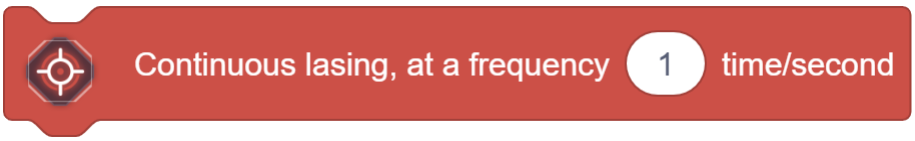
Input	Meaning	Allowed	Default
first direction	Where to look for an obstacle: Front, Back, Left, Right	dropdown	Front
second direction	Where to move if one is found: Top, Bottom, Front, Back, Left, Right	dropdown	Top
cm	How far to move	5–500 cm	50
attachment slot	Light or lasing during the move	optional	empty

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** no
- **Notes:** first run **Turn on obstacle avoidance** (Sensor), or nothing is detected. Sensors see 30–50 cm. To keep watching during a longer program, put this inside a **forever** loop. Test it with a whiteboard or a large sheet of paper.

Laser

Red blocks that fire the front laser (red, 640 nm). Another drone's laser receiver can detect the hit, which is how battles work. Never aim at eyes.

Continuous lasing

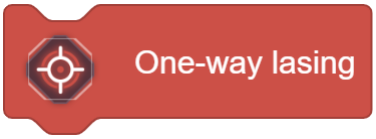


Starts firing the laser over and over until a **Stop emitting laser light** block runs.

Input	Meaning	Allowed	Default
time/second	Shots per second	1–5	1

- **Shape:** Command
- **Gives back:** nothing
- **Notes:** the program carries on to the next block while the laser keeps firing. Always pair it with **Stop emitting laser light**.

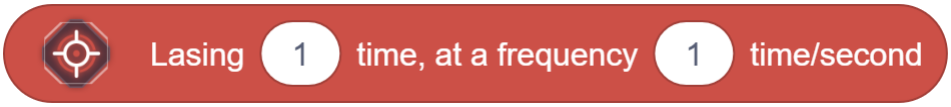
One-way lasing



Fires a single laser shot.

- **Shape:** Command
- **Inputs:** none
- **Gives back:** nothing
- **Notes:** "one-way" means one shot.

Lasing (attachment)

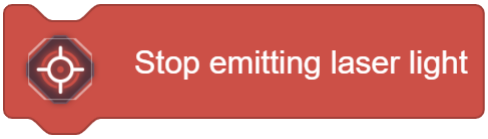


The oval laser block. It fits only in a flight block's white attachment slot and fires while that move runs.

Input	Meaning	Allowed	Default
time	Number of shots	1–20	1
time/second	Shots per second	1–5	1

- **Shape:** Value (attachment)
- **Gives back:** a laser setting for the flight block it sits in
- **Notes:** firing stops when the move ends, even if not all shots were fired. Example: **Turn to** Clockwise 360° with **Lasing** 10 times at 5 per second fires while spinning.

Stop emitting laser light



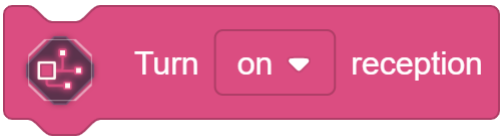
Stops a **Continuous lasing** that is running.

- **Shape:** Command
- **Inputs:** none
- **Gives back:** nothing

Sensor

Pink blocks. The first three switch sensors on and off. The value blocks read the drone's sensors while it flies. The last five drive add-on kits (electromagnet and servo) and do nothing without them. None of these work in SIM run.

Turn on/off reception

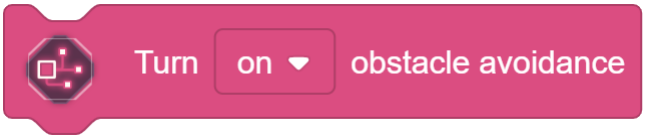


Switches the drone's laser receiver on or off. When it is off, hits from other drones' lasers are not detected.

Input	Meaning	Allowed	Default
on/off	Receiver on or off	dropdown	on

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** no

Turn on/off obstacle avoidance

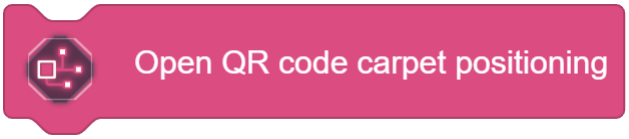


Switches the four infrared obstacle sensors (front, back, left, right) on or off. In the Programming lab they start **off**.

Input	Meaning	Allowed	Default
on/off	Sensors on or off	dropdown	on

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** no
- **Notes:** every obstacle block needs this first. The sensors see about 30–50 cm. Don't cover the light dome on top, because the receivers sit under it.

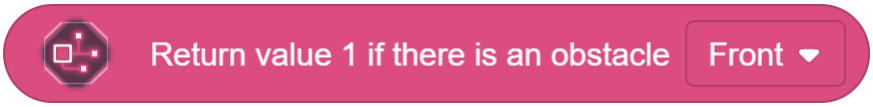
Open QR code carpet positioning



Switches the drone to positioning over a QR code mat. **Fly straight to coordinates** and **Arc flight** then use fixed positions on the mat (measured from the mat's origin) instead of distances from where the drone is.

- **Shape:** Command
- **Inputs:** none
- **Gives back:** nothing
- **SIM run:** no
- **Notes:** it turns itself off when the drone lands. QR-code alignment blocks don't work while it is on.

Return value 1 if there is an obstacle

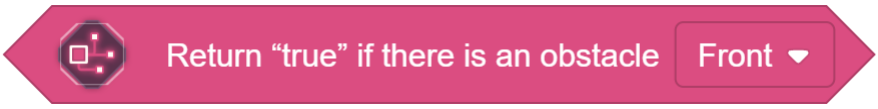


Checks one direction for an obstacle.

Input	Meaning	Allowed	Default
direction	Front, Back, Left, Right	dropdown	Front

- **Shape:** Value
- **Gives back:** 1 if there is an obstacle, 0 if not
- **SIM run:** no
- **Notes:** use it with **=** in an **if**, or to count obstacles in a variable. Needs **Turn on obstacle avoidance**.

Return “true” if there is an obstacle

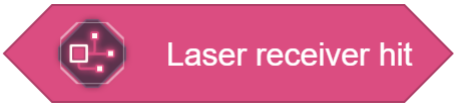


The same check, as a true/false value.

Input	Meaning	Allowed	Default
direction	Front, Back, Left, Right	dropdown	Front

- **Shape:** True/false
- **Gives back:** true if there is an obstacle, false if not
- **SIM run:** no
- **Notes:** the easier one for students. It drops straight into **if**, **wait until** or **repeat until**. To avoid missed detections, check often (in a loop) and add a short **wait**.

Laser receiver hit



Reports whether another drone's laser has hit this drone's receiver.

- **Shape:** True/false
- **Inputs:** none
- **Gives back:** true when hit, false otherwise
- **SIM run:** no
- **Notes:** needs **Turn on reception**. Good for "flash red when hit" programs.

Get angle

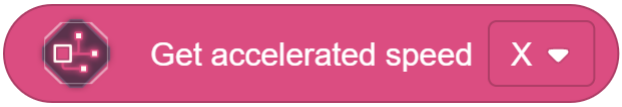


Reads the drone's current tilt or heading in degrees.

Input	Meaning	Allowed	Default
angle	Yaw (heading), Pitch (nose up/down), Roll (tilt left/right)	dropdown	Yaw

- **Shape:** Value
- **Gives back:** angle in degrees
- **SIM run:** no
- **Notes:** after turning 90° anticlockwise, yaw reads −90. **Turn to** rejects negative numbers, so wrap it in **abs** (Operators) before using it there.

Get accelerated speed

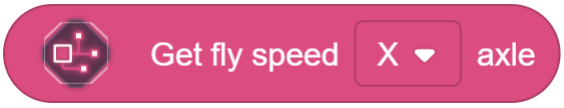


Reads the drone's acceleration along one axis.

Input	Meaning	Allowed	Default
axis	X (left/right), Y (forward/back), Z (up/down)	dropdown	X

- **Shape:** Value
- **Gives back:** acceleration on that axis (the app does not state the unit)
- **SIM run:** no
- **Notes:** have students store it in a variable and watch the numbers first, then choose a threshold.

Get fly speed



Reads how fast the drone is moving along one axis.

Input	Meaning	Allowed	Default
axis	X, Y or Z	dropdown	X

- **Shape:** Value
- **Gives back:** speed on that axis (the app does not state the unit)
- **SIM run:** no

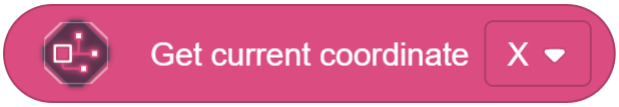
Get battery volume



Reads the battery level. "Volume" here means level.

- **Shape:** Value
- **Inputs:** none
- **Gives back:** battery percentage, 0–100
- **SIM run:** no
- **Notes:** a good safety check at the start of a program: **if** battery **<** 30 **then** Landing.

Get current coordinate



Reads where the drone is, in cm from the takeoff point.

Input	Meaning	Allowed	Default
axis	X, Y or Z	dropdown	X

- **Shape:** Value
- **Gives back:** position on that axis in cm
- **SIM run:** no
- **Notes:** positive Y is the way the nose pointed at power-on. Never test with **=** (for example Y = 200). The reading almost never hits an exact number, so use **>** or **<**.

Gets the TOF height

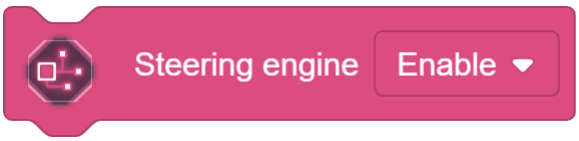


Reads the height above the floor from the downward laser distance (ToF) sensor.

- **Shape:** Value
- **Inputs:** none
- **Gives back:** height in cm
- **SIM run:** no

- **Notes:** unlike **Get current coordinate** Z, this follows the floor. Flying over a table makes it drop. Use **<** or **>**, not **=**.

Steering engine Enable/Disable (add-on)



Powers the servo add-on on or off. "Steering engine" is the app's name for a servo motor, used by the clamp kit.

Input	Meaning	Allowed	Default
state	Enable or Disable	dropdown	Enable

- **Shape:** Command
- **Gives back:** nothing
- **Notes:** needs the clamp/servo kit.

Electromagnet (attachment, add-on)

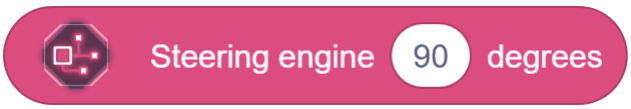


Oval version for a flight block's attachment slot. It switches the electromagnet kit during that move, for example to drop an item while flying.

Input	Meaning	Allowed	Default
state	pull (hold) or disconnect (release)	dropdown	disconnect

- **Shape:** Value (attachment)
- **Notes:** needs the electromagnet kit.

Steering engine degrees (attachment, add-on)

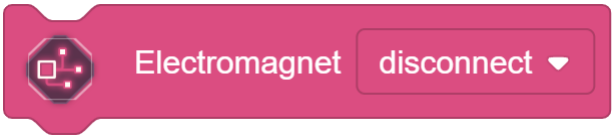


Oval version for a flight block's attachment slot. It sets the servo angle during that move, to open or close the clamp in flight.

Input	Meaning	Allowed	Default
degrees	Servo angle	0–180	90

- **Shape:** Value (attachment)
- **Notes:** needs the clamp/servo kit.

Electromagnet (command, add-on)

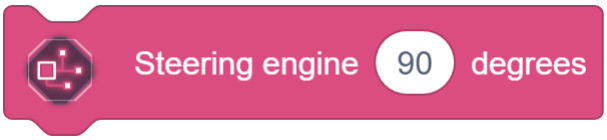


Switches the electromagnet kit while the drone is on the ground, to pick up or put down an item before takeoff or after landing.

Input	Meaning	Allowed	Default
state	pull or disconnect	dropdown	disconnect

- **Shape:** Command
- **Notes:** needs the electromagnet kit.

Steering engine degrees (command, add-on)



Sets the servo angle while the drone is on the ground.

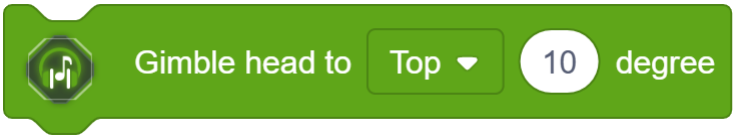
Input	Meaning	Allowed	Default
degrees	Servo angle	0–180	90

- **Shape:** Command
- **Notes:** needs the clamp/servo kit.

Multimedia

Green blocks for the front camera. "Gimble" is the app's spelling of gimbal, the motor that tilts the camera. Photos and videos are saved on the drone; download them from **Album** on the home screen. None of these work in SIM run.

Gimble head to



Tilts the front camera up or down.

Input	Meaning	Allowed	Default
direction	Top (tilt up) or Bottom (tilt down)	dropdown	Top
degree	How far to tilt	the box accepts up to 360; the camera stops at its own limit (about 130° of travel)	10

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** no
- **Notes:** run **Gimble reset** first so you tilt from level. Tilt down about 90° to film the floor.

Gimble reset



Returns the camera to level (pointing straight ahead).

- **Shape:** Command
- **Inputs:** none
- **Gives back:** nothing
- **SIM run:** no

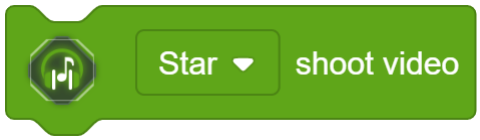
Shoot



Takes one photo (1920 × 1080) with the front camera.

- **Shape:** Command
- **Inputs:** none
- **Gives back:** nothing
- **SIM run:** no
- **Notes:** works without the live video turned on. Add **Hovering** 1 s before it so the photo isn't blurred.

Start/Stop shoot video



Starts or stops video recording (720p, 30 fps). "Star" is the app's typo for "Start".

Input	Meaning	Allowed	Default
state	Star (start) or Stop	dropdown	Star

- **Shape:** Command
- **Gives back:** nothing
- **SIM run:** no
- **Notes:** always add a **Stop** block later, or the drone keeps recording. A recording is capped at 8 minutes.

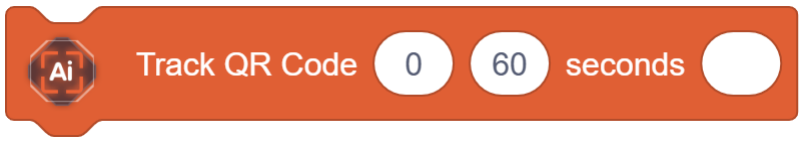
AI-Recognition

Orange blocks that use the cameras to find things. None of them work in SIM run.

Some run on the drone itself (QR codes, tag cards, lines, colour). Others run in the app on your computer (postures, images, drawing boxes). The ones that run in the app need the live video turned on first with the **image transmission** button in the Programming lab.

Every recognition needs good, even light, no strong light behind the target, a plain background, and the drone hovering 1–2 m high (or sitting on the ground). Keep the target 1–2.5 m away. Hold cards by the edge; a finger on the white border stops recognition.

Track QR Code

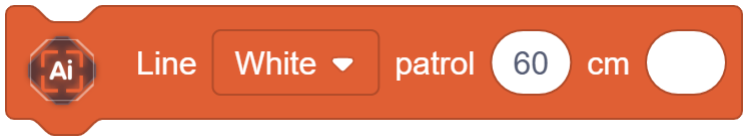


The front camera locks onto a numbered QR code and the drone follows it, keeping the distance it started at, for a number of seconds.

Input	Meaning	Allowed	Default
first number	Which QR code to follow	0–9	0
seconds	How long to follow	0–600	60
attachment slot	Light or lasing while tracking	optional	empty

- **Shape:** Command
- **Gives back:** nothing
- **Notes:** the program waits here until the time is up. If no code is seen for 30 s it gives up ("Search timeout"). Move the code slowly, under 0.3 m/s sideways. Fly 1–2.2 m high. Video drops to 360p while tracking.

Line patrol

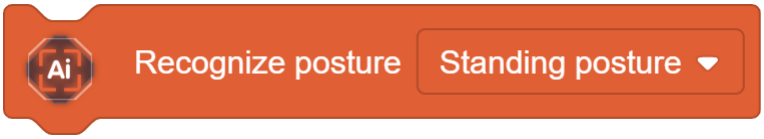


The downward camera follows a line on the floor for a set distance.

Input	Meaning	Allowed	Default
colour	White = light line on a dark floor; Black = dark line on a light floor	dropdown	White
cm	Distance to follow, about ±10 cm	0–600	60
attachment slot	Light or lasing while following	optional	empty

- **Shape:** Command
- **Gives back:** nothing
- **Notes:** lines at least 5 cm wide, high contrast, no shiny tape. Turns must be gentle: the angle between line segments must be over 135°, so the nose turns at most 45° at a corner. If no line is seen for 30 s it stops ("Line patrol timeout"). Fly 1–2.2 m high.

Recognize posture

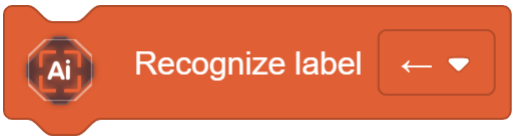


Looks for a person in a chosen pose for up to 5 seconds.

Input	Meaning	Allowed	Default
posture	Standing posture, Sitting posture, Ultraman ray, Raise your hands above your head, Raise your hands to your shoulders, Extend your left arm, Stretch out your right arm	dropdown	Standing posture

- **Shape:** Command
- **Gives back:** nothing directly. Read the result with **Return the result of Recognize**.
- **Runs in:** the app. Turn on image transmission first.
- **Notes:** for standing, keep hands below the waist and feet in the picture. For sitting, keep feet in the picture.

Recognize label

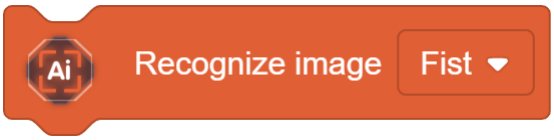


Looks for a printed tag card for up to 5 seconds.

Input	Meaning	Allowed	Default
label	Arrows ← → ↑ ↓, digits 0–9, letters A–Z	dropdown	←

- **Shape:** Command
- **Gives back:** nothing directly. Use **Return the result of Recognize**.
- **Runs on:** the drone
- **Notes:** arrow cards are good for "follow the signs" tracks. Number cards are good for picking a route or collecting items in a game.

Recognize image

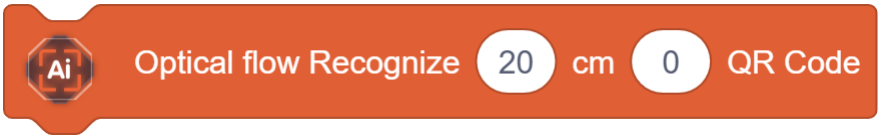


Looks for a type of object for up to 5 seconds.

Input	Meaning	Allowed	Default
image	Fist, Hand, Pedestrian, Hula (another Hula drone)	dropdown	Fist

- **Shape:** Command
- **Gives back:** nothing directly. Use **Return the result of Recognize**.
- **Runs in:** the app. Turn on image transmission first.
- **Notes:** another Hula is recognised best from an angle, not straight on.

Optical flow Recognize QR Code

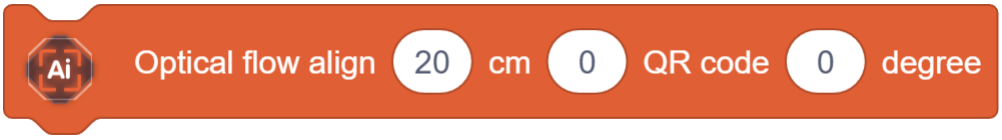


The downward camera looks for a numbered QR code on the floor.

Input	Meaning	Allowed	Default
cm	Printed size of the QR code	6–30 cm in practice	20
second number	Which QR code	0–9	0

- **Shape:** Command
- **Gives back:** nothing directly. Use **Return the result of Recognize**, and **Get the coordinate … relative to the tag card** for where it is.
- **Runs on:** the drone
- **Notes:** the code must be almost directly under the drone. At 1 m height that means within about 30 cm; fly higher to see further.

Optical flow align QR code

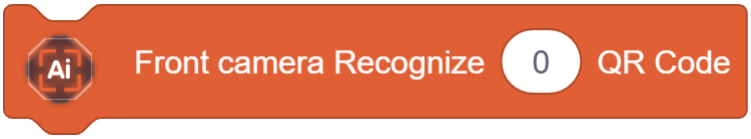


Centres the drone over a floor QR code, lines the nose up with the code, then turns by an extra angle.

Input	Meaning	Allowed	Default
cm	Printed size of the QR code	6–30 cm in practice	20
second number	Which QR code	0–9	0
degree	Extra turn after aligning	−180 to 180	0

- **Shape:** Command
- **Gives back:** nothing
- **Runs on:** the drone
- **Notes:** follow it with **Landing** for a precise landing, or use **Land on QR code**, which does both. Keep no more than 3 codes in view at once.

Front camera Recognize QR Code

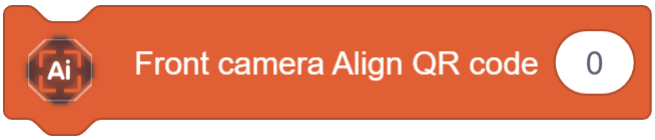


The front camera looks for a numbered QR code in front of the drone, for example on a wall.

Input	Meaning	Allowed	Default
number	Which QR code	0–9	0

- **Shape:** Command
- **Gives back:** nothing directly. Use **Return the result of Recognize**.
- **Runs on:** the drone
- **Notes:** keep the code 1–2.2 m from the camera, with the camera level.

Front camera Align QR code



Turns the nose and slides sideways a little to bring a wall QR code to the centre of the camera view.

Input	Meaning	Allowed	Default
number	Which QR code	0–9	0

- **Shape:** Command
- **Gives back:** nothing
- **Runs on:** the drone
- **Notes:** each run lasts only 0.3 s, so put it in a **repeat** loop (for example repeat 20). This is how to aim the laser at a target: align, then **One-way lasing**.

Return the result of Recognize



Tells you whether the last recognition block succeeded.

- **Shape:** Value
- **Inputs:** none
- **Gives back:** 1 if found, 0 if not
- **Notes:** only the most recent recognition counts. Typical use: **if** (Return the result of Recognize **=** 1) **then** …

Return a RGB of Color Recognition

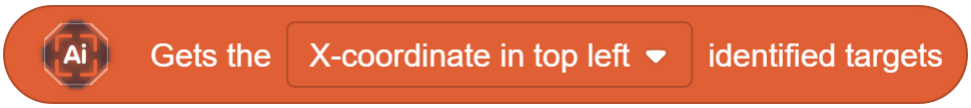


The front camera finds the biggest coloured area in view (it ignores black, white and grey) and reports its colour.

- **Shape:** Value
- **Inputs:** none
- **Gives back:** an RGB colour value
- **Runs on:** the drone

- **Notes:** drop it into the colour slot of a **Light - Color value** block to make the drone glow the colour it sees, or add it to a list. You can't do maths with it. Bright, matt, strongly coloured cards work best. Turn the drone's own light off first so it doesn't tint the view.

Gets the coordinate of identified targets

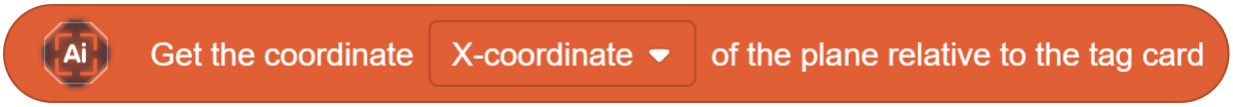


Gives one corner of the box around the last thing recognised, in pixels on the video image.

Input	Meaning	Allowed	Default
corner	X or Y of the top-left corner, or X or Y of the bottom-right corner	dropdown	X top left

- **Shape:** Value
- **Gives back:** a pixel position on the video image
- **Notes:** use the four values with **Coordinate drawing box** to draw a box around the target. If several targets are seen, it reports the largest.

Get the coordinate relative to the tag card

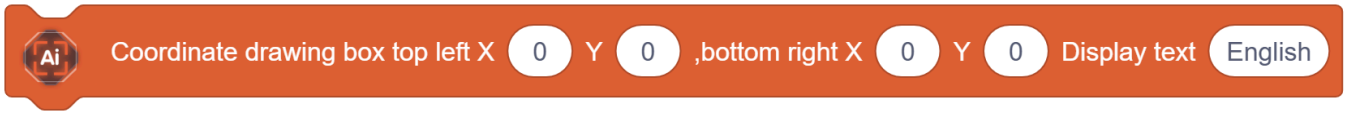


After a QR code, number or arrow card is recognised, gives where that card is compared with the drone.

Input	Meaning	Allowed	Default
value	X-coordinate, Y-coordinate, Z-coordinate (cm) or Angle (degrees)	dropdown	X-coordinate

- **Shape:** Value
- **Gives back:** distance in cm, or angle in degrees
- **Runs on:** the drone
- **Notes:** floor cards are measured with the downward camera, wall cards with the front camera. Feed the values into **Fly straight to coordinates** to fly to the card. Doesn't work while QR carpet positioning is on.

Coordinate drawing box



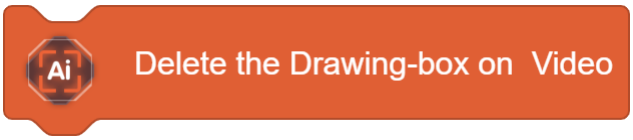
Draws a rectangle with a text label on the live video in the app.

Input	Meaning	Allowed	Default
top left X, Y	Top-left corner in pixels	0–600	0
bottom right X, Y	Bottom-right corner in pixels	0–600	0
Display text	Label shown with the box	any text	English

- **Shape:** Command

- **Gives back:** nothing
- **Runs in:** the app. Turn on image transmission first.
- **Notes:** usually filled with four **Gets the coordinate of identified targets** blocks, so the box follows what was recognised.

Delete the Drawing-box on Video



Removes the boxes drawn on the video.

- **Shape:** Command
- **Inputs:** none
- **Gives back:** nothing
- **Runs in:** the app

Control

Green blocks that decide what runs, when, and how often. They are standard Scratch blocks and all work in SIM run.

wait seconds

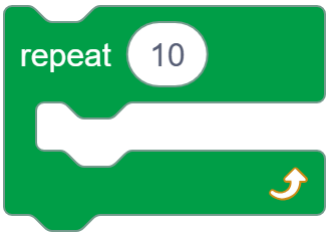


Pauses the script for a number of seconds before the next block.

Input	Meaning	Allowed	Default
seconds	How long to pause	any number, decimals allowed	1

- **Shape:** Command
- **Gives back:** nothing
- **Notes:** in the air, the drone simply holds position while the script waits. Useful between light blocks so each colour shows.

repeat



Runs the blocks inside it a set number of times.

Input	Meaning	Allowed	Default
number	How many times	whole number	10

- **Shape:** Wrapper
- **Gives back:** nothing
- **Notes:** a square is **repeat** 4 with **Fly to** Front + **Turn to** 90° inside.

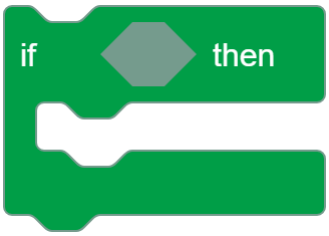
forever



Runs the blocks inside it again and again until the program is stopped.

- **Shape:** Wrapper (nothing can go below it)
- **Inputs:** none
- **Gives back:** nothing
- **Notes:** with flight blocks inside, the drone keeps flying until the battery runs low or you press Stop. Usually used in a second script (started by a broadcast) to keep checking a sensor.

if then

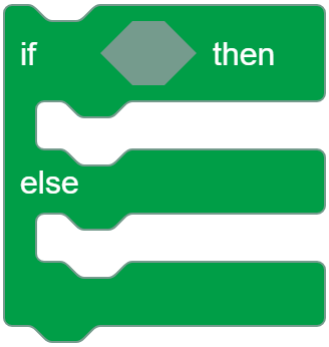


Runs the blocks inside only if the condition is true.

Input	Meaning	Allowed	Default
hexagon slot	The condition	a true/false block	empty

- **Shape:** Wrapper
- **Gives back:** nothing
- **Notes:** it checks once, when it is reached. To keep checking, put it inside **forever** or **repeat**.

if then else



Runs the top group if the condition is true, otherwise the bottom group.

Input	Meaning	Allowed	Default
hexagon slot	The condition	a true/false block	empty

- **Shape:** Wrapper
- **Gives back:** nothing

wait until

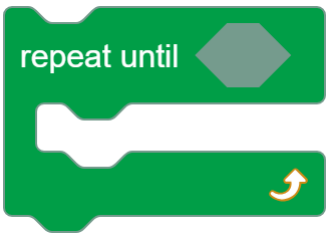


Pauses the script until the condition becomes true.

Input	Meaning	Allowed	Default
hexagon slot	The condition	a true/false block	empty

- **Shape:** Command
- **Gives back:** nothing
- **Notes:** example: **wait until** obstacle in Front, then **Landing**. The drone hovers while it waits.

repeat until



Runs the blocks inside again and again until the condition becomes true.

Input	Meaning	Allowed	Default
hexagon slot	The condition that ends the loop	a true/false block	empty

- **Shape:** Wrapper

- **Gives back:** nothing
- **Notes:** example: **Fly to** Back 10 cm at 20% **repeat until** there is no obstacle in Front.

stop



Stops scripts that are running.

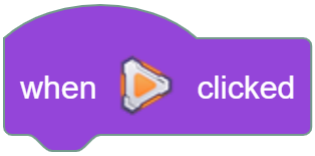
Input	Meaning	Allowed	Default
what	all = every script; this script = only the one this block is in; other scripts = every script except this one	dropdown	all

- **Shape:** Command
- **Gives back:** nothing
- **Notes:** to stop a **forever** laser running in another script, use **stop all**.

Events

Purple blocks that start scripts and let scripts signal each other. A program can have several scripts running at the same time; a broadcast is how one starts another.

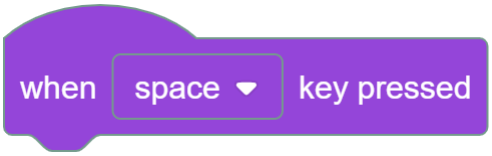
when clicked (start)



Starts the script below it when the program is run (with the Run or SIM run button).

- **Shape:** Start block
- **Inputs:** none
- **Gives back:** nothing
- **Notes:** every program needs at least one. Blocks not connected under a start block never run, a common reason a program "does nothing".

when key pressed



Starts the script below it each time a keyboard key is pressed.

Input	Meaning	Allowed	Default
key	Which key	space, arrow keys, any, a–z, 0–9	space

- **Shape:** Start block
- **Gives back:** nothing
- **Notes:** PC keyboard only. Handy for a "panic" script: **when** space **key pressed** → **Landing**.

when I receive

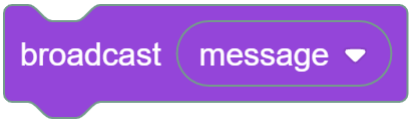


Starts the script below it when a matching broadcast message is sent.

Input	Meaning	Allowed	Default
message	Which message to listen for. Pick "New message" to create one.	dropdown	message

- **Shape:** Start block
- **Gives back:** nothing
- **Notes:** a broadcast interrupts flight. A new flight command from the receiving script replaces the one in progress.

broadcast

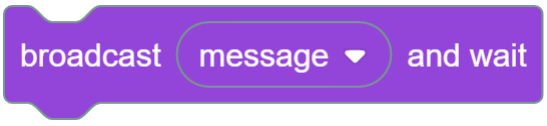


Sends a message, which starts every **when I receive** script for that message, and carries on at once.

Input	Meaning	Allowed	Default
message	Which message to send	dropdown	message

- **Shape:** Command
- **Gives back:** nothing
- **Notes:** typical use: after **Take off**, broadcast "watch" to start a **forever** loop that checks obstacles while the main script flies the route.

broadcast and wait



Sends a message, then waits until every script it started has finished before going on.

Input	Meaning	Allowed	Default
message	Which message to send	dropdown	message

- **Shape:** Command
- **Gives back:** nothing
- **Notes:** if a receiving script contains **forever**, this never continues.

Operators

Light-green blocks for maths, comparisons and text. They never run on their own: they go inside another block's slot. All work in SIM run.

add (+)



- **Inputs:** two numbers (typed, or value blocks)
- **Shape:** Value
- **Gives back:** their sum
- **Example:** distance + 20 to fly a little further each lap.

subtract (–)



- **Inputs:** two numbers
- **Shape:** Value
- **Gives back:** first minus second

multiply (*)



- **Inputs:** two numbers
- **Shape:** Value
- **Gives back:** their product

divide (/)



- **Inputs:** two numbers
- **Shape:** Value
- **Gives back:** first divided by second (can be a decimal)
- **Notes:** a regular polygon turn is 360 / number of sides.

pick random



Input	Meaning	Default
first number	Lowest possible value	1
second number	Highest possible value	10

- **Shape:** Value
- **Gives back:** a random number between the two, inclusive
- **Notes:** two whole numbers give a whole number; if either has a decimal point, you get decimals.

greater than (>)



- **Inputs:** two values (right side defaults to 50)
- **Shape:** True/false
- **Gives back:** true if the left is bigger
- **Notes:** use this, not =, with live sensor readings.

less than (<)



- **Inputs:** two values (right side defaults to 50)
- **Shape:** True/false
- **Gives back:** true if the left is smaller

equals (=)



- **Inputs:** two values (right side defaults to 50)
- **Shape:** True/false
- **Gives back:** true if they are the same
- **Notes:** good for counters and recognition results (= 1). Bad for sensor readings, which rarely hit an exact number.

and



- **Inputs:** two true/false blocks
- **Shape:** True/false
- **Gives back:** true only if both are true
- **Example:** obstacle in Front **and** obstacle on Left.

or



- **Inputs:** two true/false blocks
- **Shape:** True/false
- **Gives back:** true if either one is true

not



- **Inputs:** one true/false block
- **Shape:** True/false
- **Gives back:** the opposite
- **Example: repeat until not** obstacle in Front.

join



- **Inputs:** two pieces of text (defaults: apple, banana)
- **Shape:** Value
- **Gives back:** the two stuck together, for example "applebanana"
- **Notes:** handy for the text on a **Coordinate drawing box**.

letter of



Input	Meaning	Default
number	Position of the letter, starting at 1	1
text	The text to look in	apple

- **Shape:** Value
- **Gives back:** that one character

length of



- **Inputs:** text (default: apple)
- **Shape:** Value

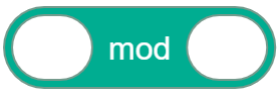
- **Gives back:** the number of characters

contains



- **Inputs:** a text (default: apple) and what to look for (default: a)
- **Shape:** True/false
- **Gives back:** true if the first text contains the second

mod



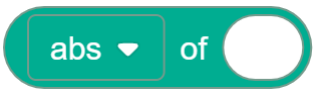
- **Inputs:** two numbers
- **Shape:** Value
- **Gives back:** the remainder after dividing, for example 10 mod 3 = 1
- **Notes:** a counter mod 2 = 0 means "every second lap", useful for alternating light colours.

round



- **Inputs:** one number
- **Shape:** Value
- **Gives back:** the nearest whole number, for example 5.8 → 6

maths function (abs of)



Input	Meaning	Default
function	abs, floor, ceiling, sqrt, sin, cos, tan, asin, acos, atan, ln, log, e ^, 10 ^	abs
number	The number to work on	empty

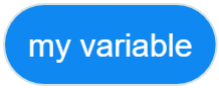
- **Shape:** Value
- **Gives back:** the result
- **Notes: abs** removes the minus sign. Use it to feed a negative yaw angle into **Turn to**.

Variables and lists

These blocks only appear after you create a variable or list. In the Variables category, click **Make a Variable** or **Make a List** and type a name. To rename or delete one, right-click its name on PC, or press and hold on a tablet.

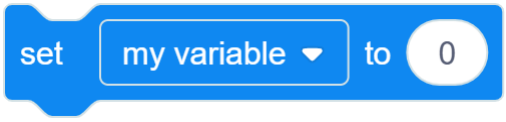
A **variable** holds one value (a number or text). A **list** holds many values in order, numbered from 1. Use descriptive names such as "laps" or "obstacle count". All of these work in SIM run.

variable (value)



- **Shape:** Value
- **Inputs:** none (the name is the block)
- **Gives back:** the variable's current value
- **Notes:** drop it into any white slot, for example the cm of **Fly to**.

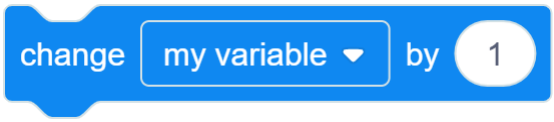
set variable to



Input	Meaning	Default
variable	Which variable	dropdown
value	New value (number, text or value block)	0

- **Shape:** Command
- **Gives back:** nothing
- **Notes:** set counters to 0 at the start of the program.

change variable by



Input	Meaning	Default
variable	Which variable	dropdown
number	Amount to add (negative to subtract)	1

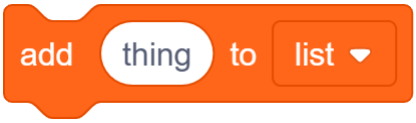
- **Shape:** Command
- **Gives back:** nothing
- **Example:** count obstacles: each time one is seen, change "obstacle count" by 1. On the 3rd, fire the laser.

list (value)



- **Shape:** Value
- **Gives back:** the whole list as one piece of text

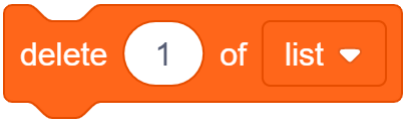
add to list



Input	Meaning	Default
thing	Value to add	thing
list	Which list	dropdown

- **Shape:** Command
- **Notes:** adds to the end. Example: add **Return a RGB of Color Recognition** five times to "learn" five colours.

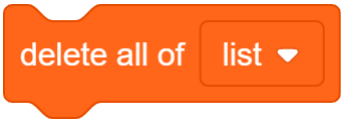
delete of list



Input	Meaning	Default
number	Position to remove	1
list	Which list	dropdown

- **Shape:** Command
- **Notes:** later items move up one place.

delete all of list



- **Inputs:** which list
- **Shape:** Command
- **Notes:** empty the list at the start of a program so old values don't mix in.

insert at list

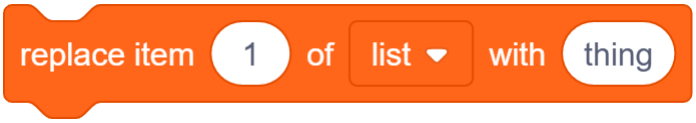


Input	Meaning	Default
thing	Value to insert	thing
number	Position	1
list	Which list	dropdown

- **Shape:** Command

- **Notes:** items from that position on move down one place.

replace item of list



Input	Meaning	Default
number	Position to replace	1
list	Which list	dropdown
thing	New value	thing

- **Shape:** Command

item of list



Input	Meaning	Default
number	Position	1
list	Which list	dropdown

- **Shape:** Value
- **Gives back:** the value at that position
- **Example:** a list of distances (100, 50, 100, 50) and **Fly to** Front using **item** *lap* **of** distances draws a rectangle.

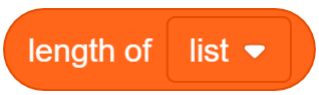
item # of in list



Input	Meaning	Default
thing	Value to find	thing
list	Which list	dropdown

- **Shape:** Value
- **Gives back:** the position of the first match, or 0 if it isn't there

length of list



- **Inputs:** which list
- **Shape:** Value

- **Gives back:** how many items are in the list
- **Notes:** use as the count in **repeat** to go through every item.

list contains



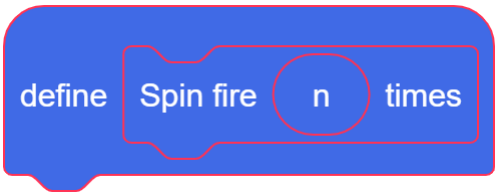
Input	Meaning	Default
list	Which list	dropdown
thing	Value to look for	thing

- **Shape:** True/false
- **Gives back:** true if the value is anywhere in the list

My Blocks

My Blocks lets students package a group of blocks under one name and reuse it, with inputs, like a function. Click **Make a Block**, type a name, and use **Add an input** for each value it needs. The example below is a block named "Spin fire *n* times" with one input, *n*.

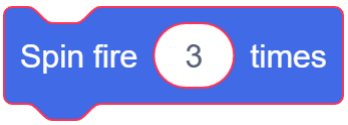
define



The start block for your own block. Snap under it the blocks that should run when your block is used.

- **Shape:** Start block
- **Inputs:** the input names you added (here *n*). Drag the *n* oval from the define block into slots below it, for example into **repeat**.
- **Gives back:** nothing
- **Example: define** Spin fire *n* times → **repeat** *n* → **Turn to** Clockwise 90° + **One-way lasing**.

your custom block



Appears in the My Blocks palette once defined. Using it runs the blocks under its **define**.

- **Shape:** Command
- **Inputs:** one slot per input you added (here 3 is passed in as *n*)
- **Gives back:** nothing
- **Notes:** a good way to teach decomposition. Each leg or trick of a track becomes one named block, and the main script becomes short and readable.

Battle mode

Laser-tag between drones. In Battle mode the drones' lights and obstacle avoidance are off.

Shared rules: each drone has 300 HP. A laser hit removes 100. At 0 HP the drone is "defeated": it can't fire or be hit until it respawns after a short wait.

Battle controls: fire button (one laser shot per tap), gimbal control (aim up/down), HP bar, and a crosshair showing where the laser points.

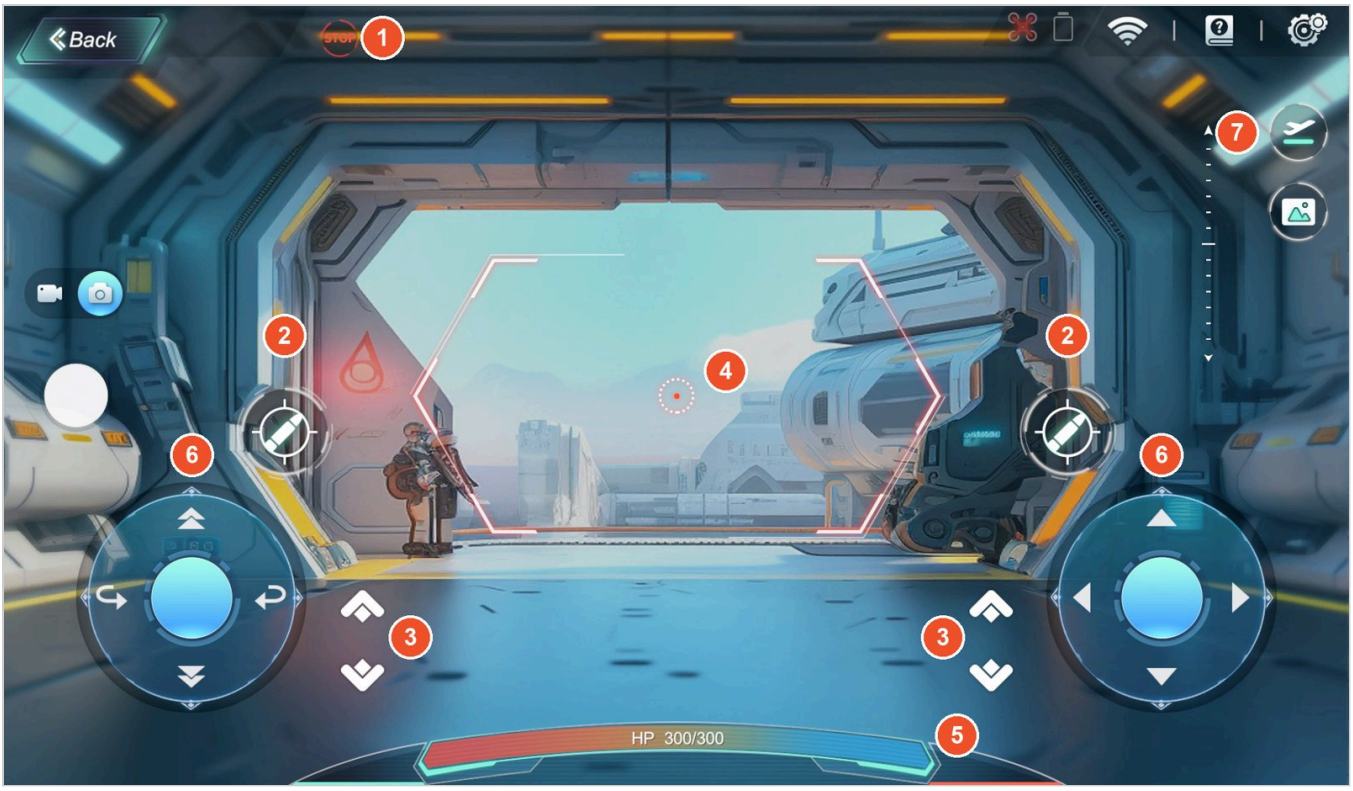
Calibrate the crosshair first, or students miss even when aiming well. Open **Settings** inside the battle screen, aim at a white wall, fire, tap the red laser dot on screen, then tap **X**.

Version 2.0.5 has four battle modes. They are not what the official manual describes, because the manual was written for an older version with "Melee mode" and "Scrimmage".



Mode	What we found	Connection
Scrimmage	Opens the battle flying screen straight away: free-for-all, like the manual's "Melee mode"	Direct (AP) mode, one drone per player
Team play	Red vs Blue team game with a room owner, like the manual's "Scrimmage". It wouldn't open without drones on a router.	STATION (router) mode
Stronghold	Not tested (needs drones)	Probably router mode
Battlefield	Not tested (needs drones)	Probably router mode

The battle flying screen



#	Control	What it does
1	**STOP**	Emergency stop
2	Fire buttons (left and right)	Fire one laser shot per tap, from either thumb
3	Gimbal up/down arrows	Tilt the camera to aim higher or lower
4	Crosshair	Where the laser will hit (calibrate it first)
5	HP bar	Health, starting at 300
6	Joysticks	Fly, as on the Flight screen
7	Takeoff / Landing	As on the Flight screen

Running a team game (manual's "Scrimmage", probably "Team play" in 2.0.5)

1. Put all drones and devices on one router (see **Connecting to the drone → Router mode**).
2. The first player to open the team mode becomes the **room owner**. Everyone else waits 10 seconds, then joins.
3. The owner sets game time, target points, flight speed and number of players. Players pick a free Red or Blue slot.
4. When everyone is ready, the owner taps **Start**.
5. At the end, the results screen shows each player's kills, deaths and assists (KDA) and an MVP for each team.

Album, Profile and Settings

Album

Photos and videos are stored on the drone, not the laptop. Connect to the drone, open **Album**, and download what you want to keep. Press and hold a file to select several for deleting when the drone's storage is full.

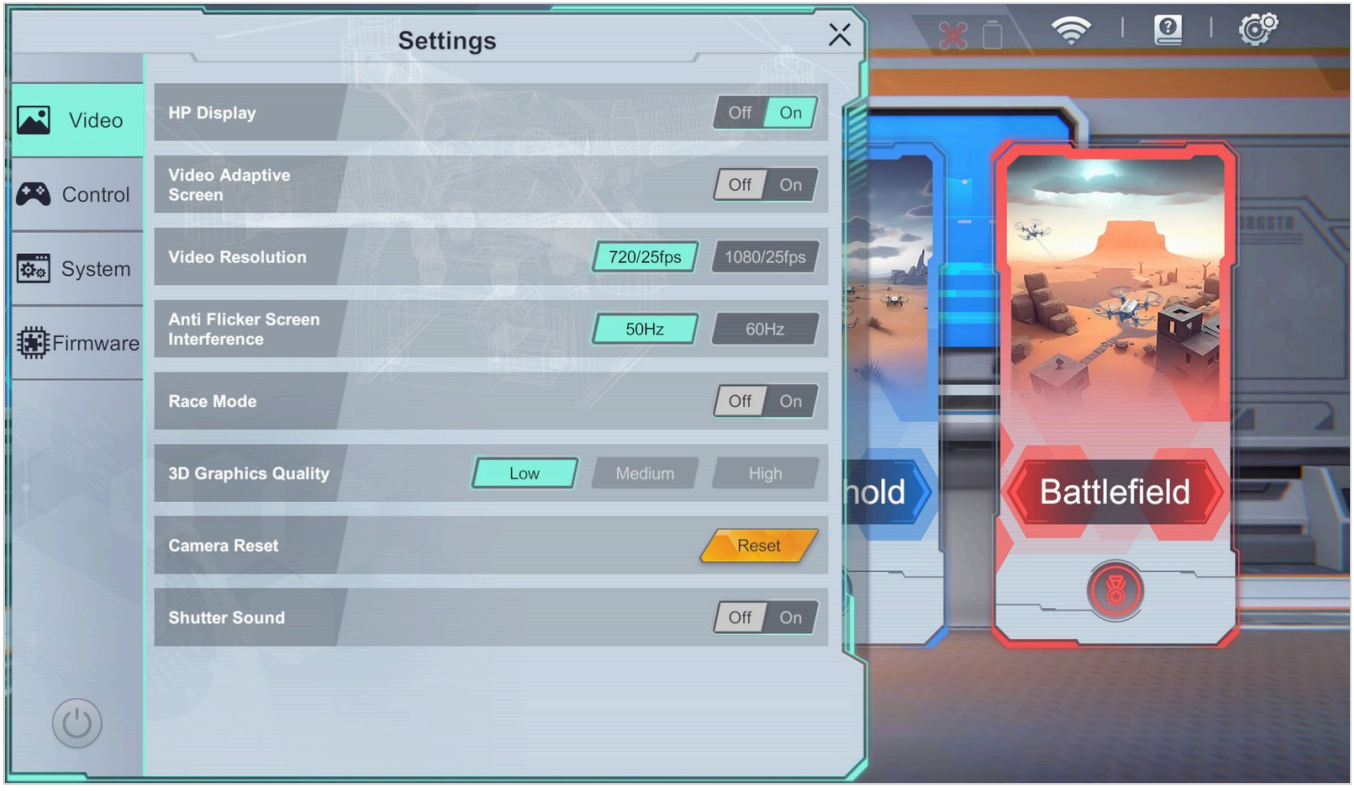
Profile

Item	What it does
Personal info	Change picture, name, gender
Password management	Change password with a verification code
Message center	System messages
Technical support	User manual, feedback, after-sales contact
Log out	Sync work to the cloud first, or it may be lost

Settings

Settings (the gear, top right) has four tabs: Video, Control, System and Firmware.

Video tab

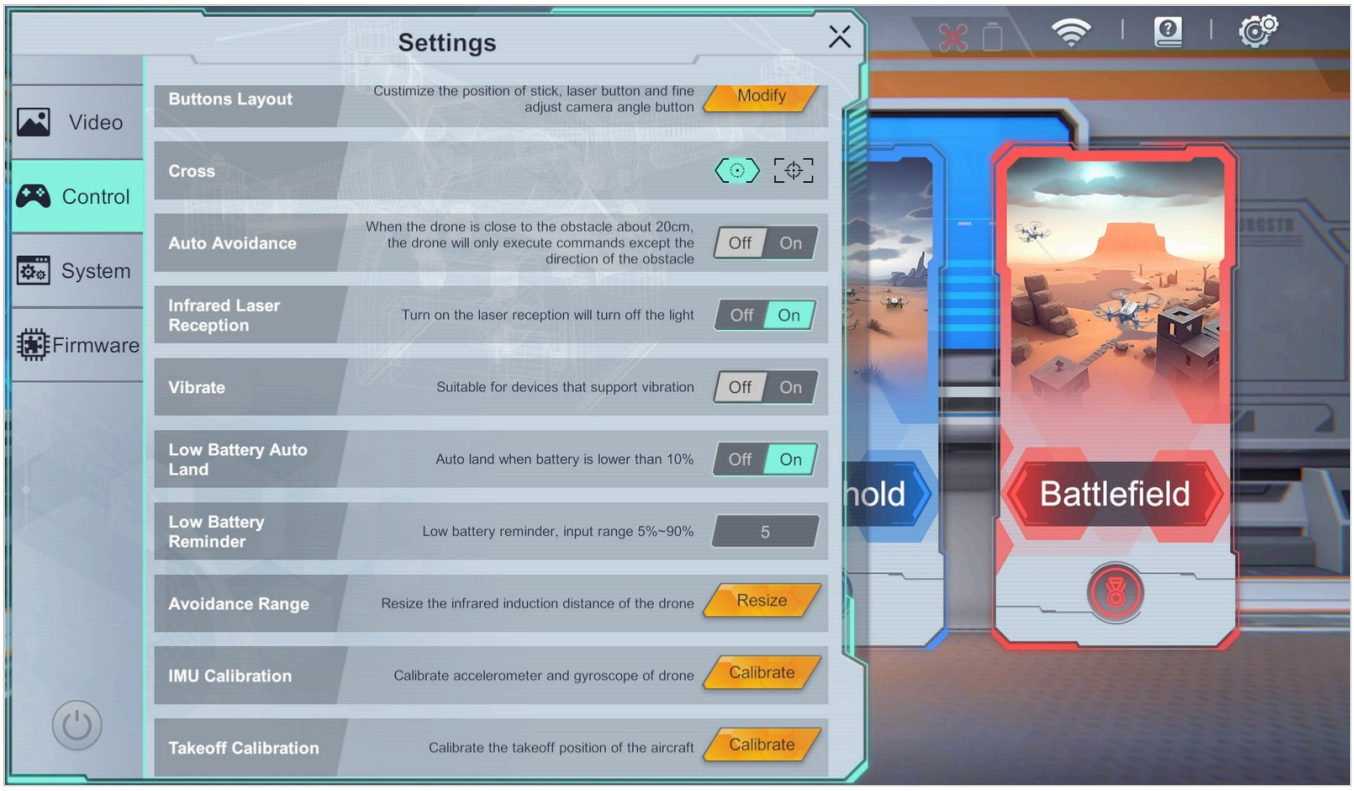


Setting	Plain meaning	Suggested
HP Display	Show health bars in battle	On

Setting	Plain meaning	Suggested
Video Adaptive Screen	Fit the camera picture to screens that aren't 16:9	On for tablets
Video Resolution	720p or 1080p, both 25 fps	720p (smoother over Wi-Fi)
Anti Flicker Screen Interference	Stops flicker from room lights: 50 Hz or 60 Hz	Match your mains power (50 Hz in most of Europe, Africa, Asia and the Middle East; 60 Hz in the Americas)
Race Mode	Makes the drone resist Wi-Fi interference when many drones fly (the manual's "Competition mode")	On for battles and competitions
3D Graphics Quality	Low / Medium / High graphics for the 3D scenes and simulator	Low on older laptops
Camera Reset	Level the front camera	—
Shutter Sound	Click sound on photos	Your choice
Power button (bottom left)	Turn off the connected drone while it is landed	—

Control



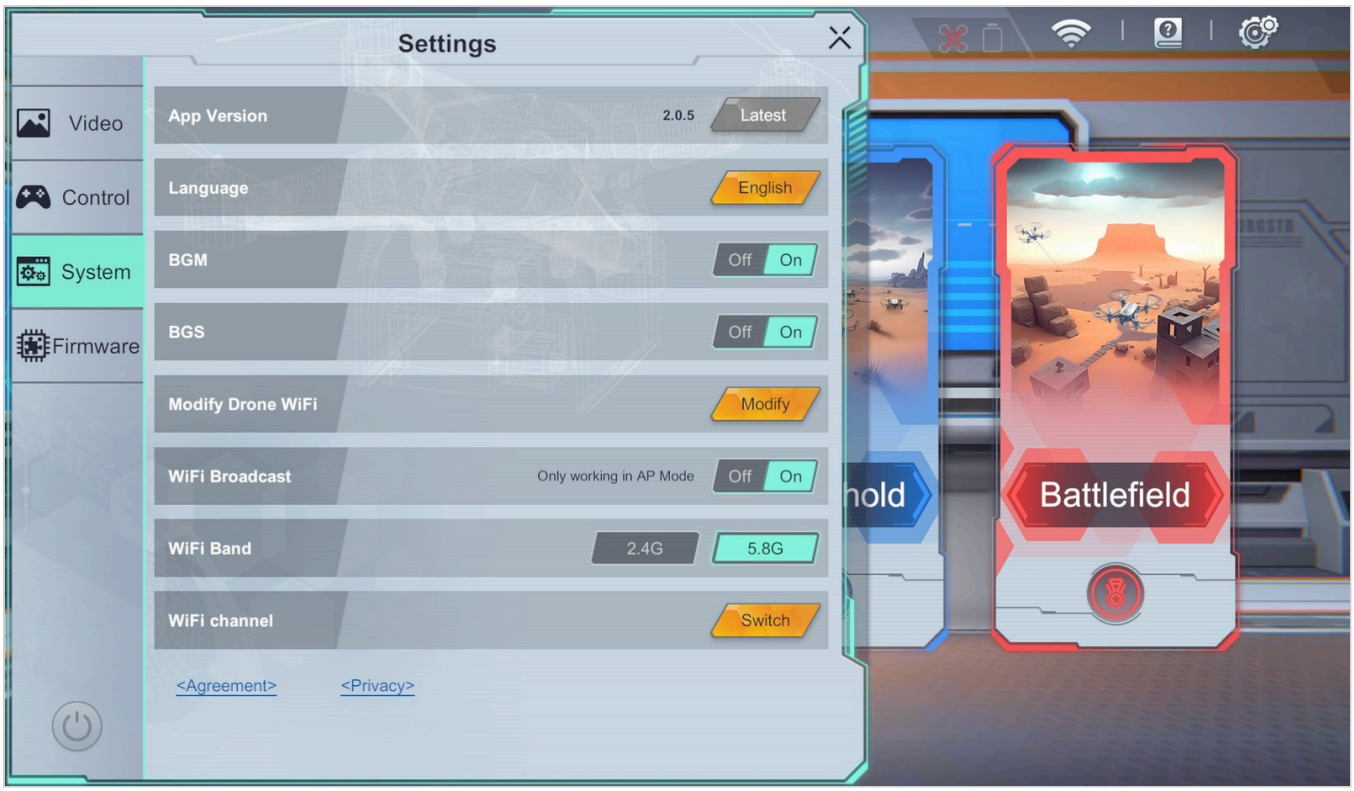


Setting	Plain meaning	Suggested
Light	Top light brightness: Off, Low, Medium, High	Medium
Move Speed	Joystick flying speed: Low, Medium, High (about 0.5, 1 and 2 m/s)	Low for beginners
Landing Speed	How fast it comes down when landing: Low or High	Low
Track Angle Sensitivity	Turning speed: Low, Medium, High (about 45, 90 and 120 °/s)	Low for beginners
Control Mode	Joystick layout: Japanese hand or American hand	American hand (default)
InputDevice	PC controls: Keyboard+Mouse, or Keyboard only	—
Dynamic Stick	The joystick appears where your thumb touches	On for tablets
Buttons Layout	**Modify** moves the joysticks, laser button and camera-angle buttons on screen	—
Cross	Crosshair style for battles (two choices)	—
Auto Avoidance	When the drone is about 20 cm from an obstacle, it ignores commands that would fly it into the obstacle	On for beginners
Infrared Laser Reception	Detect laser hits. The app notes that turning it on turns the drone's light off.	On for battles
Vibrate	Tablet vibrates when hit	—
Low Battery Auto Land	Lands by itself under 10% battery	**Keep on**
Low Battery Reminder	Warning level, 5–90% (default 5)	30
Avoidance Range	**Resize** adjusts how far the infrared obstacle sensors see	Leave unless faulty
IMU Calibration	Recalibrate the motion sensors (accelerometer and gyroscope)	Only if the drone tilts or drifts

Setting	Plain meaning	Suggested
Takeoff Calibration	Calibrate the takeoff position	Only if takeoffs drift

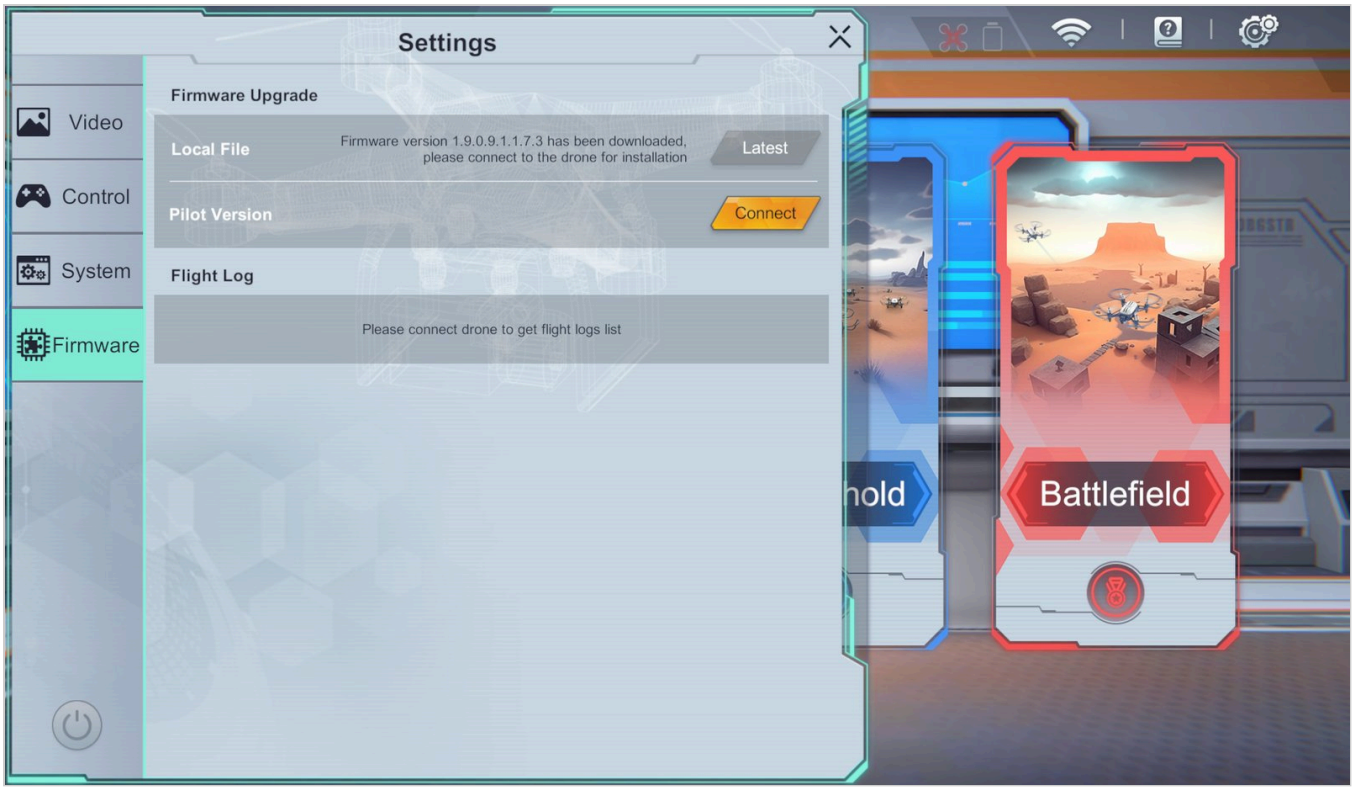
These settings apply to manual flying and battles, not to the Programming lab.

System



Setting	Plain meaning
App Version	Installed version (2.0.5). The button says **Latest**, or offers an update when one exists.
Language	App language. Changing it can switch account server (see Installing).
BGM	Background music on/off
BGS	Sound effects on/off
Modify Drone WiFi	Rename the drone's Wi-Fi or change its password, or send it a router's details. The drone restarts.
WiFi Broadcast	Remote ID broadcast required by some countries' drone rules. Works only in AP mode. Leave on.
WiFi Band	Switch the drone between 2.4G and 5.8G. Rejoin with the password afterwards. 5.8 GHz often works better in crowded classrooms.
WiFi channel	**Switch** moves the drone to another Wi-Fi channel. Try it when drones interfere with each other.

Firmware



Setting	Plain meaning
Local File	Firmware already downloaded to this computer, for example "Firmware version 1.9.0.9.1.1.7.3 has been downloaded, please connect to the drone for installation". **Latest** means there is nothing newer to download.
Pilot Version	The firmware on the drone. Shows **Connect** until a drone is connected, then offers the upgrade.
Flight Log	The drone's flight logs, listed after connecting (useful for support)

Troubleshooting

Problem	Likely cause	Fix
Drone won't take off, or flips over on takeoff	Propellers on the wrong motors	**75 mmR** blades front-left and back-right, **75 mm** blades front-right and back-left
Program does nothing	Blocks not attached under **when clicked**, or **Running** was tapped instead of held	Snap the stack under the start block; press and hold **Running**
Takes off then just hovers	No **Landing** at the end, or a **forever** loop	Add **Landing**; press **Stop** to land now
Drifts sideways while hovering	Floor too plain, shiny or dark; dirty downward camera; poor light	Use a textured mat, more light, wipe the belly camera
Moves are sloppy or overlap	No pause between moves	Add **Hovering** 1–2 s after each move; lower the speed
Several light blocks, only the last shows	Light blocks don't wait	Put **Hovering** or **wait** between them
Number won't change when typed	Outside the block's allowed range	See the range in the block's entry
Obstacle blocks never react	Obstacle avoidance is off in the Programming lab	Add **Turn on obstacle avoidance** after takeoff
Posture / image recognition always fails	Image transmission is off (these run in the app)	Turn on image transmission before running
QR code not found	Wrong number or size typed; code too far or not under the drone; finger on the border; glare	Check the number and cm; 1 m height = within 30 cm below; hold by the edge
ToF / optical flow red before takeoff	Normal on the ground	Take off; they turn green
Can't find the Hula Wi-Fi	Drone in router mode (white flash)	Press the power button 3 times for direct mode (purple flash)
Connected to the wrong drone	Neighbouring drone has a similar name	Check the serial on the label; number drones and laptops
Lots of dropouts with many drones	Crowded 2.4 GHz Wi-Fi	Switch drones to 5.8 GHz (Settings → System); turn on Race Mode (Settings → Video)
Short flights	Old or cold battery, lights on, heavy tricks	Charge fully; replace after about 30 h of flying; fly above 5 °C
Firmware update failed	Connection dropped mid-update	Tap **Re upgrade**; if it won't work, hold reset 10 s to roll back
Forgot a changed drone Wi-Fi password	—	Hold reset 5 s: back to Hula-XXXXXXXX / 12345678
Lost students' programs	Saved only on one laptop, or overwritten by Sync	Upload to cloud before Sync; Export programs as backup

Repairs: contact your dealer, or HighGreat support at service@hg-fly.com.

Glossary of app wording

The app says	It means
Aircraft, plane	The drone
Single driver	Manual flying with the on-screen joystick
Flight order	Flight blocks ("order" = command)
Program lab, Single device programming	The block programming editor
SIM run	Run on the 3D simulator, not the real drone
Image transmission	Live video from the drone's camera
Gimble, gimbal, PTZ, head	The motor that tilts the front camera
Steering engine, steering gear	Servo motor (add-on clamp kit)
Grab kit	Clamp add-on
Lasing, one-way lasing	Firing the laser; one shot
Reception	The laser receiver
Crush out	Light off
Colorful lantern	Light cycling through colours
Breathing light	Light slowly fading in and out
Star	Start (typo on the video block)
Battery volume	Battery level (%)
Accelerated speed	Acceleration
Fly speed … axle	Speed along an axis
TOF height	Height above the floor from the downward distance sensor
Optical flow	The downward camera, used to hold position and read floor codes
Front camera, forward shooting, proactive	The front camera
Tag card, label	Printed arrow, number or letter cards
QR code carpet, QR-code carpet positioning	A floor mat of QR codes for exact positioning
Patrol, line patrol, hunting line	Follow a line on the floor
Surround, fly around	Fly in a circle around a point
Yaw / Pitch / Roll	Heading / nose up-down / tilt left-right
Direct connection mode	Laptop joins the drone's own Wi-Fi (the app's Connect screen calls this AP Mode)
STATION Mode, networking mode, router mode	Drones and laptops all on one router
Equipment base, Device	List of drones the app can see
Melee mode	Free-for-all battle (old manual name; called Scrimmage in 2.0.5)
Scrimmage	In 2.0.5: free-for-all battle. In the old manual: Red vs Blue team battle (now Team play)
Settlement	End-of-game results screen

The app says	It means
Facula	The laser dot on a wall
American hand / Japanese hand	The two joystick layouts
Flight control firmware	The drone's software